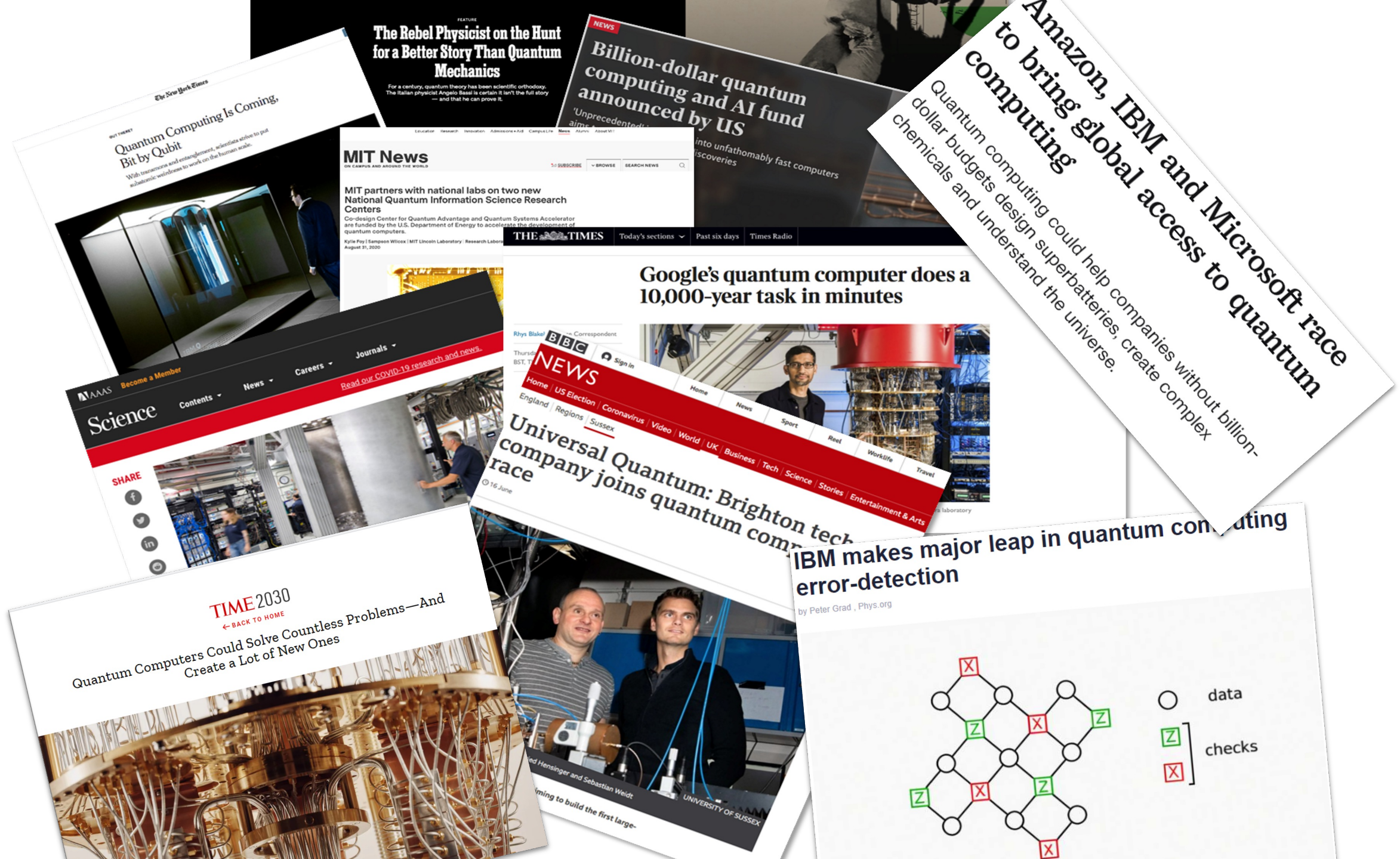


Cloud-based Quantum Computing 101

Giuseppe Bisicchia and Antonio Brogi

Department of Computer Science
University of Pisa

Quantum Computing



Quantum Computing Is Coming, Bit by Qubit

With transistors and entanglement, scientists strive to put quantum weirdness to work on the human scale.



The Rebel Physicist on the Hunt for a Better Story Than Quantum Mechanics

For a century, quantum theory has been scientific orthodoxy. The Italian physicist Angelo Bassi is certain it isn't the full story — and that he can prove it.

MIT News

MIT partners with national labs on two new National Quantum Information Science Research Centers

Co-design Center for Quantum Advantage and Quantum Systems Accelerator are funded by the U.S. Department of Energy to accelerate the development of quantum computers.

Kylie Fay | Sampson Wilcox | MIT Lincoln Laboratory | Research Laboratory August 31, 2020

Billion-dollar quantum computing and AI fund announced by US

'Unprecedented' aims to speed up quantum computing into unfathomably fast computers

THE NEW YORK TIMES Today's sections Past six days Times Radio

Amazon, IBM and Microsoft race to bring global access to quantum computing

Quantum computing could help companies without billion-dollar budgets design superbatteries, create complex chemicals and understand the universe.

Google's quantum computer does a 10,000-year task in minutes

NEWS

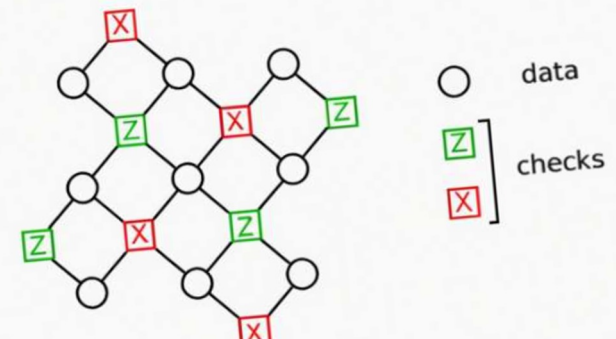
Home US Election Coronavirus Video World UK Business Tech Science Stories Entertainment & Arts

Universal Quantum: Brighton tech company joins quantum computing race

© 16 June

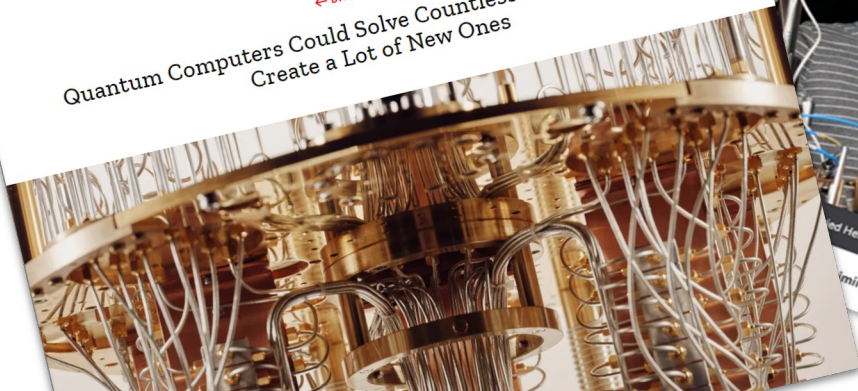
IBM makes major leap in quantum computing error-detection

by Peter Grad, Phys.org

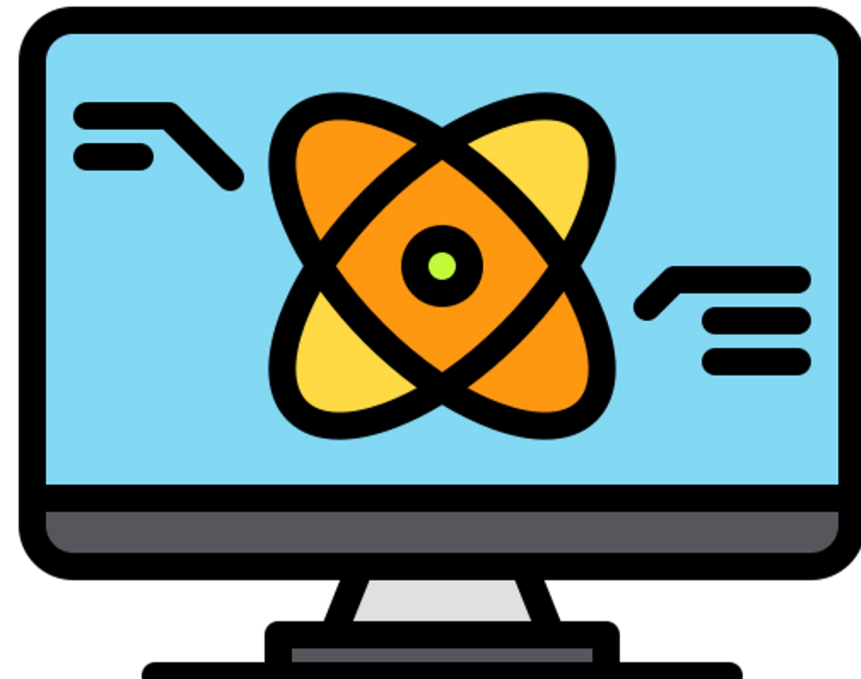


TIME 2030

Quantum Computers Could Solve Countless Problems—And Create a Lot of New Ones



Quantum Computing Fundamentals





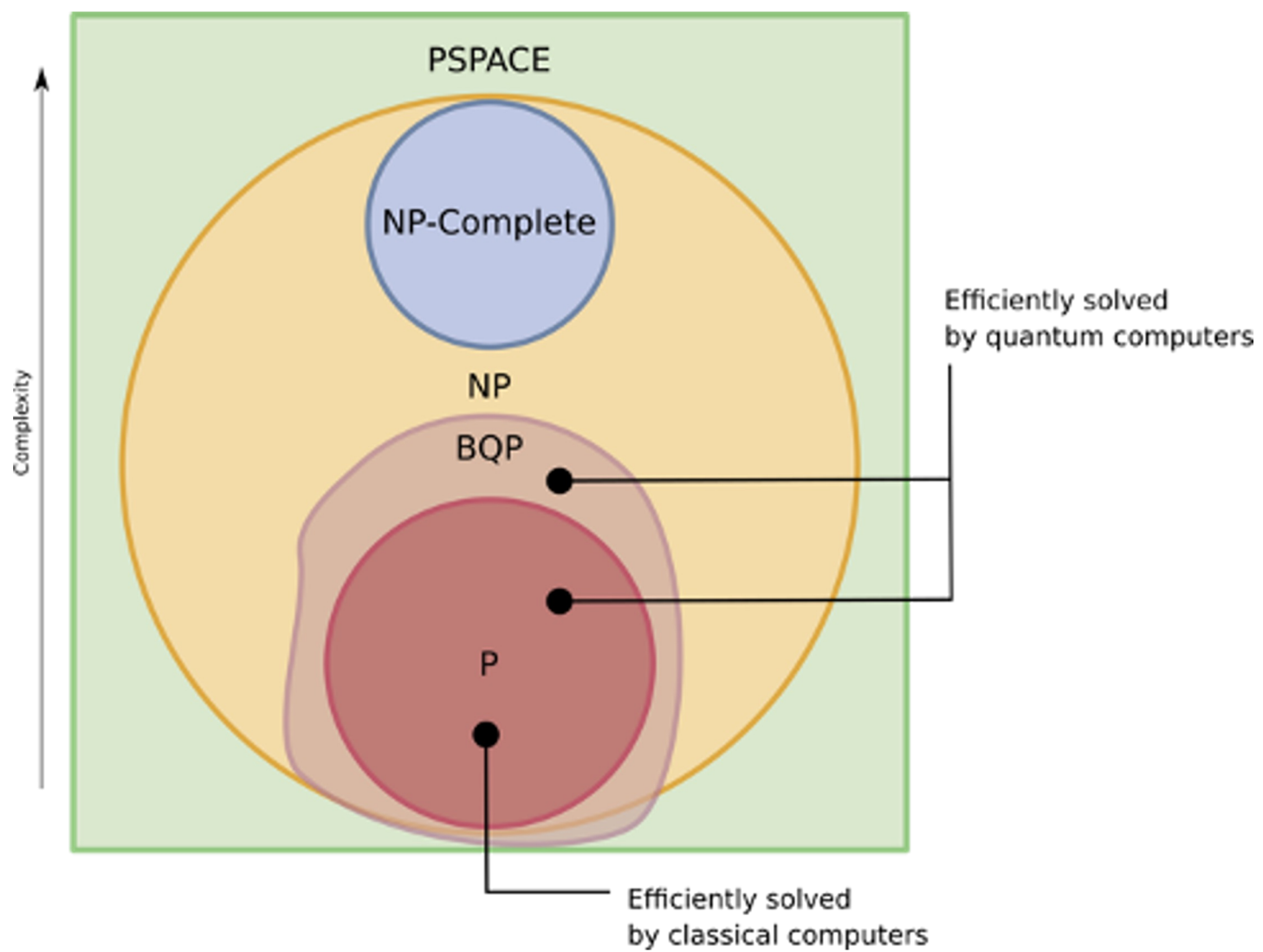
*Why should I invest in
Quantum
Computing?*

G

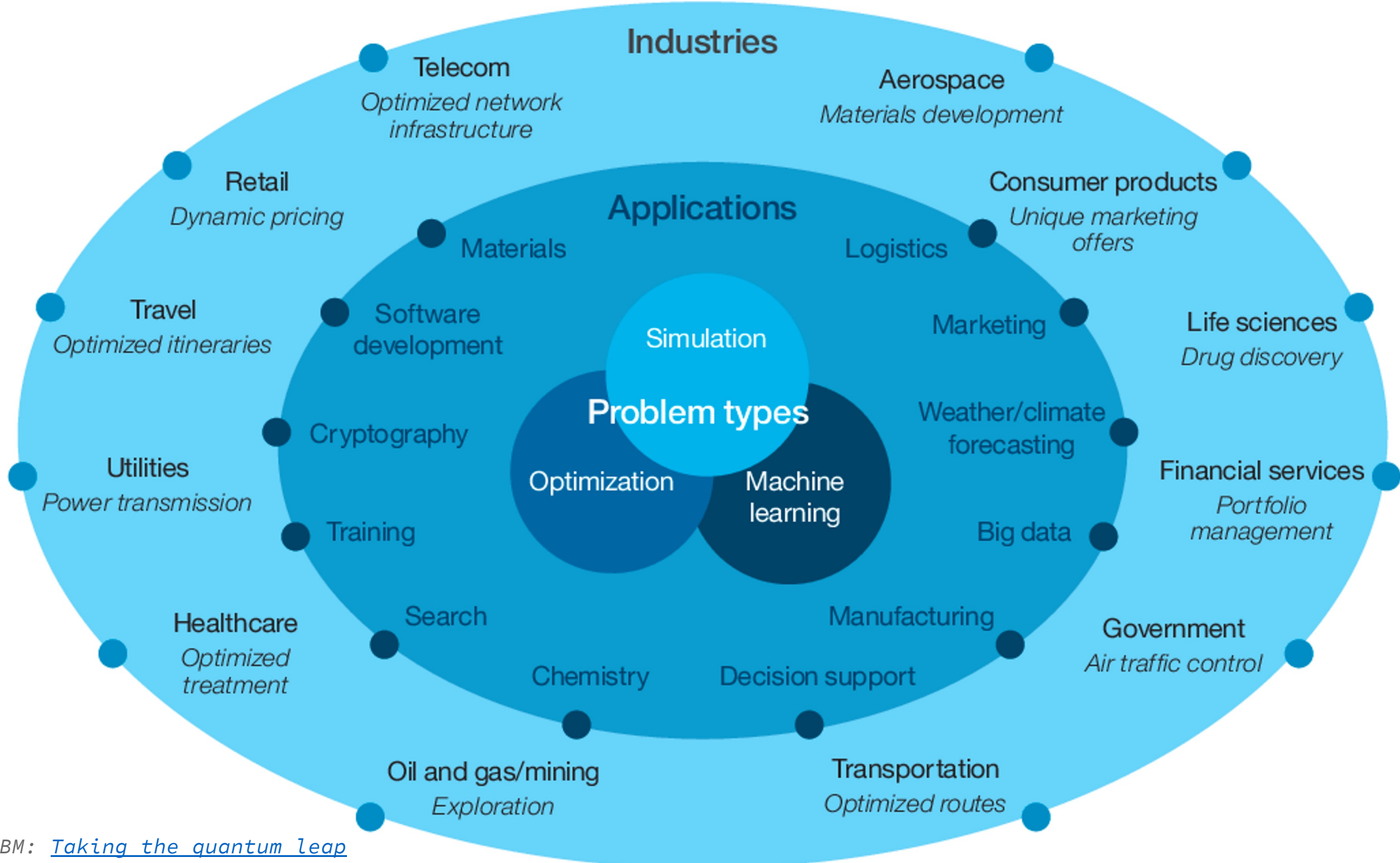
Why should I invest in Quantum Computing?



Investing in quantum computing is a strategic move for those with a long-term perspective. Quantum computing has the potential to revolutionize various industries by solving complex problems exponentially faster than classical computers. It promises breakthroughs in fields like cryptography, drug discovery, materials science, and optimization.



Problems	Explication	Quantum Complexity	Classical Complexity
Factorization (Shor)	Decomposition of a number into a product of smaller integers	$O(\log N)$	$\Theta\left(\exp\left(\left(\frac{32}{9}n\right)^{\frac{1}{3}}(\log n)^{\frac{2}{3}}\right)\right)$
Search (Grover)	Search in an unordered sequence of data	$O(\sqrt{n})$	$O(n)$



Quantum Computing

Quantum computers can solve various problems **exponentially faster** than traditional computers

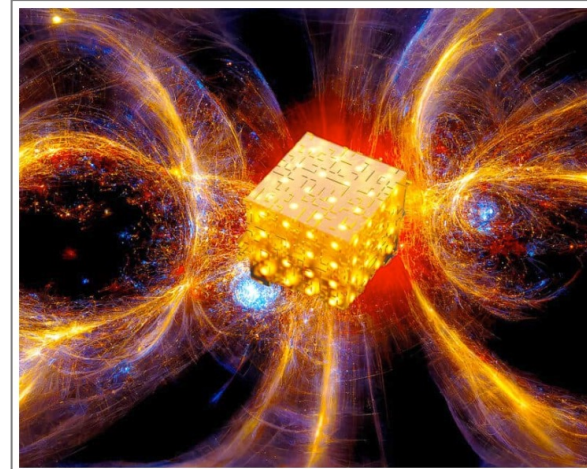
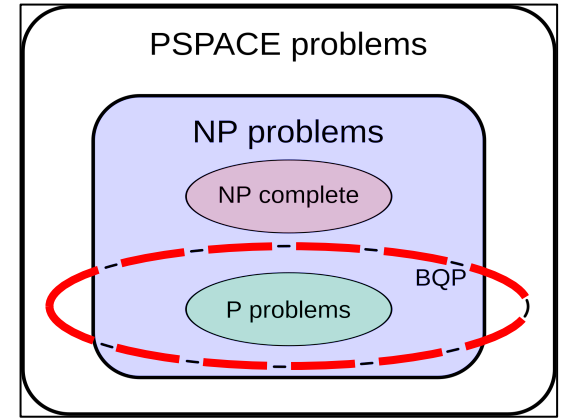
15 Significant Ways Quantum Computing Could Soon Impact Society



Expert Panel® Forbes Councils Member

Apr 18, 2023,

1. Breaking Current Encryption Schemes
2. Better Securing Sensitive Data
3. 'Breaking' The Blockchain
4. Modeling Chemical Reactions For Drug Development
5. Enhancing Drug Discovery And Personalized Medicine
6. Improving AI Capabilities
7. Optimizing Investment Portfolios
8. Safe Computing Of Encrypted Data
9. Democratizing Generative AI
10. Enabling True Real-Time Reporting
11. Discovering New Materials
12. Improving Weather Forecasting
13. Enabling Hyper-Personalized Shopping Experiences
14. Optimizing Traffic Flows
15. Combating Climate Change



Quantum computer built by Google can instantly execute a task that would normally take 47 years

*What is a Quantum
Computer?*

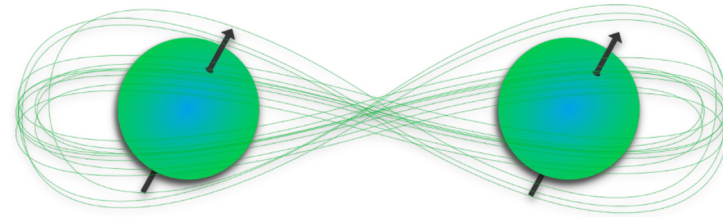


What is a Quantum Computer?

A Quantum Computer is a device that leverages
Quantum Mechanics Properties to perform **computations**



Superposition



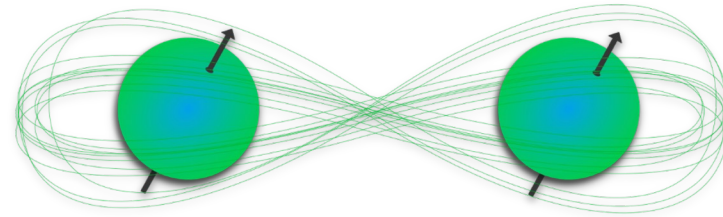
Entanglement

What is a Quantum Computer?

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Superposition



Entanglement

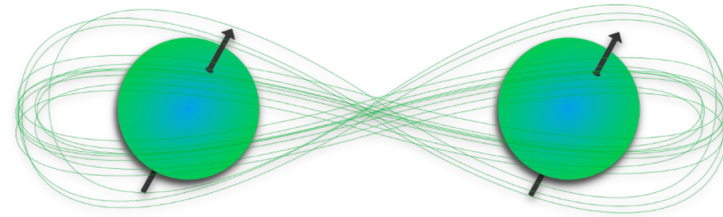
In essence, qubits leverage the principles of **quantum mechanics** to provide a **new way of representing and processing information**, promising **significant advantages** for solving specific problems when compared to classical computing

What is a Quantum Computer?

A Quantum Computer is a device that leverages
Quantum Mechanics Properties to perform **computations**



Superposition



Entanglement

From Bits...

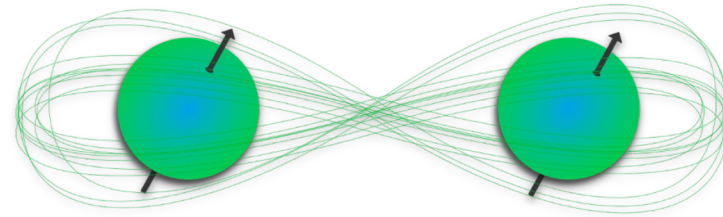


What is a Quantum Computer?

A Quantum Computer is a device that leverages
Quantum Mechanics Properties to perform **computations**



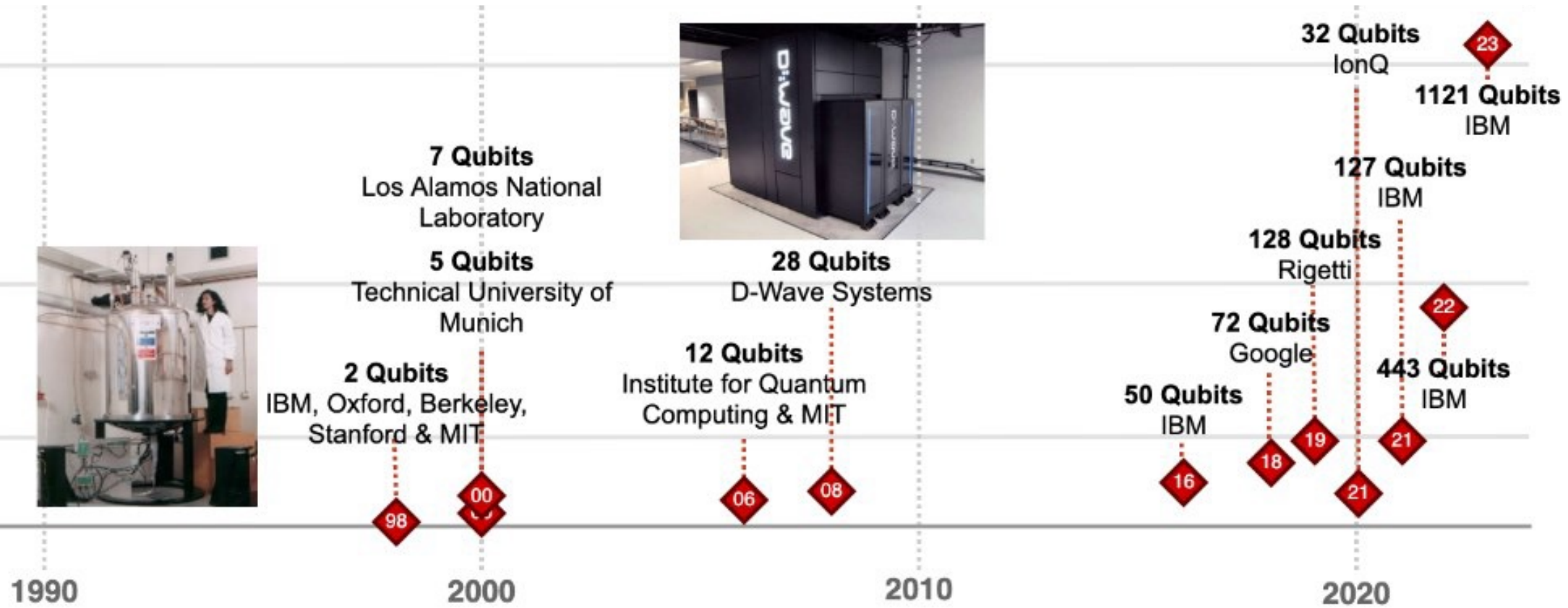
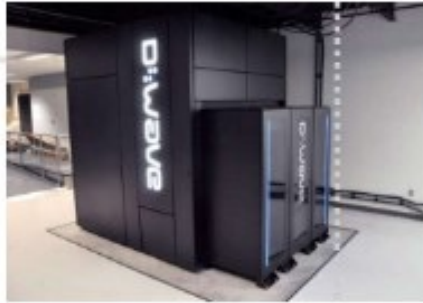
Superposition



Entanglement



...to Qubits!



A Qubit

Mathematically, a qubit is a **complex-valued linear combination** of its two possible basis states “0” (the *ground state*) and “1” (the *excited state*).

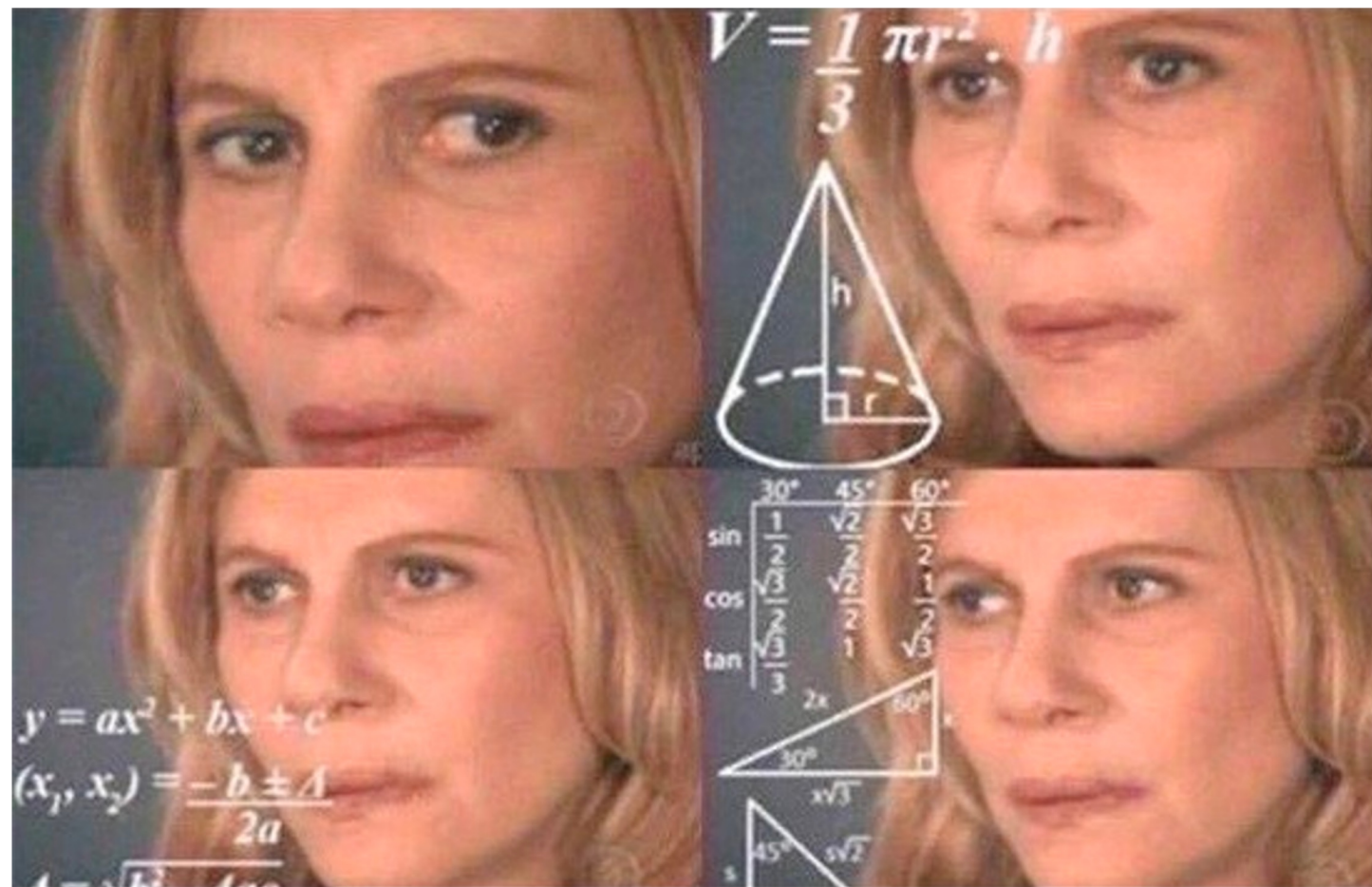
$$|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

The diagram illustrates the mathematical representation of a qubit state $|\Psi\rangle$ as a linear combination of two basis states, $|0\rangle$ and $|1\rangle$. The equation is $|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle$. The ket $|\Psi\rangle$ is enclosed in a red box and labeled "Qubit" with a red arrow. The coefficients α and β are each enclosed in a blue box and labeled "amplitudes: complex numbers" with blue arrows. The basis kets $|0\rangle$ and $|1\rangle$ are each enclosed in a red box and labeled "kets: 2-dim vectors" with red arrows. The diagram uses color-coding: red boxes and arrows for the qubit and basis states, and blue boxes and arrows for the complex-valued amplitudes.

Qubit

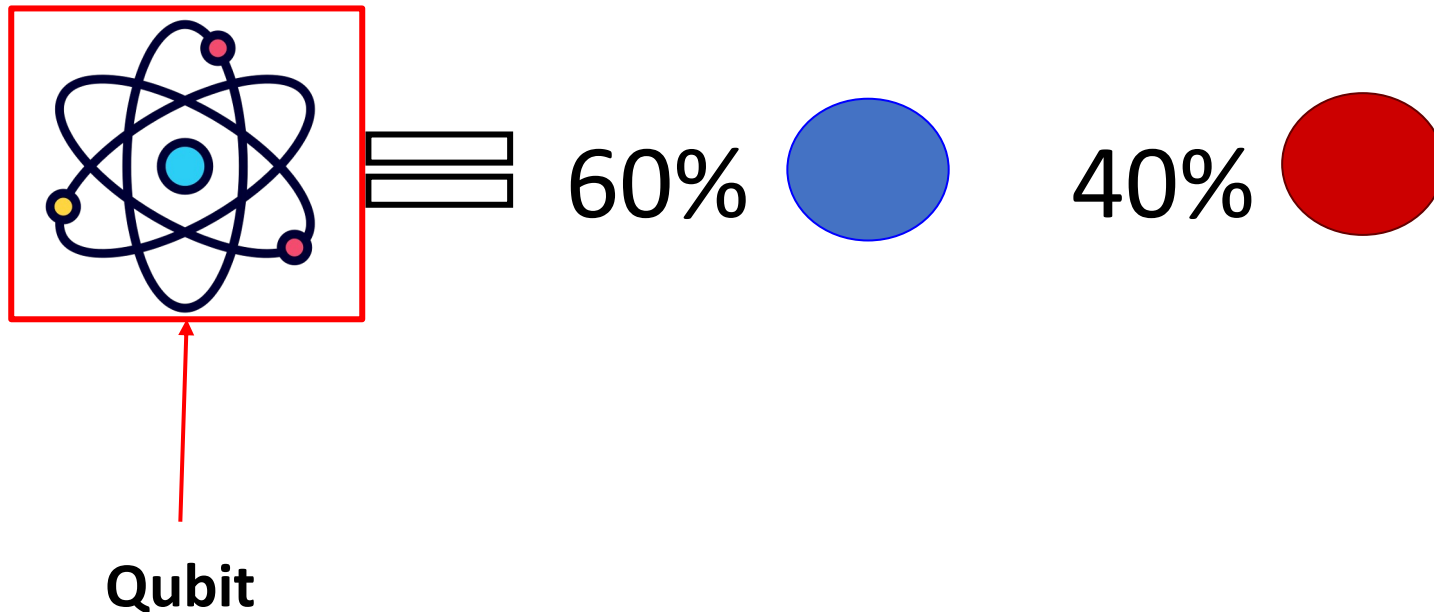
kets: 2-dim
vectors

amplitudes:
complex
numbers



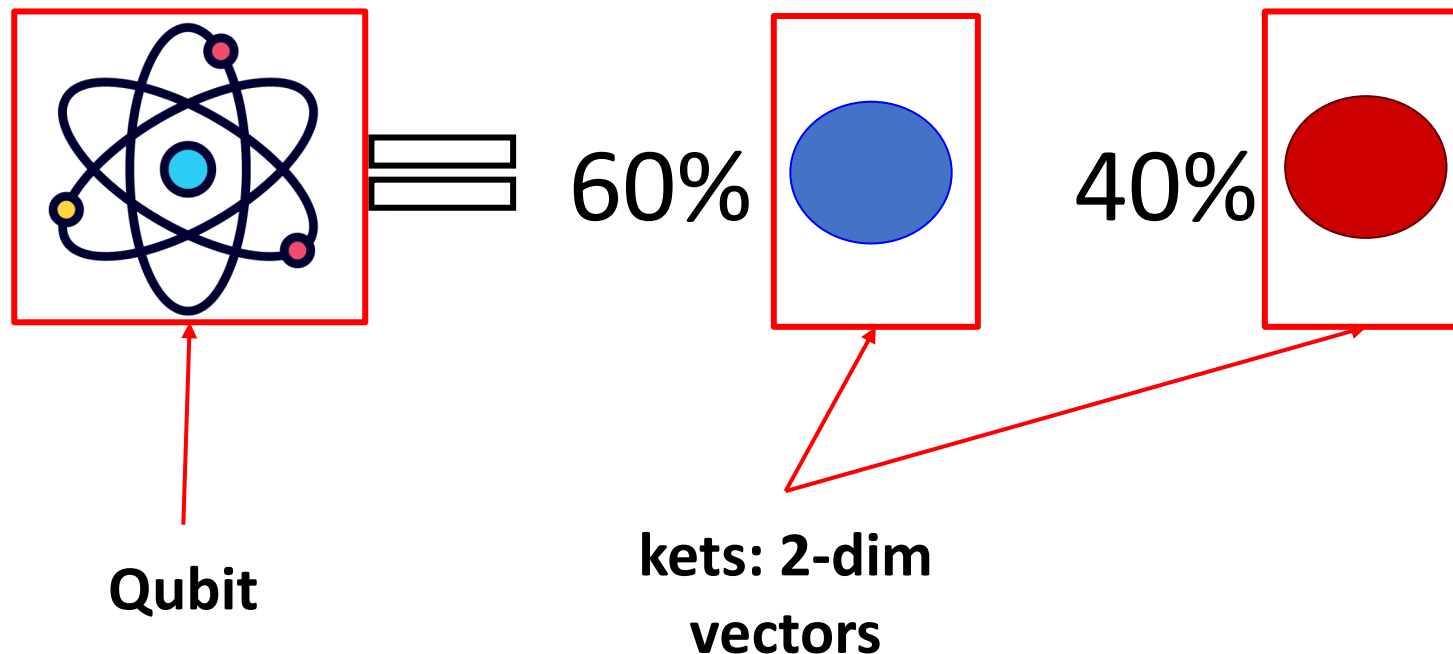
A Qubit

Conceptually, a qubit is the **fundamental unit of quantum information**. Unlike classical the bit, its state can be “0”, “1” or a ***superposition*** (i.e., linear combination) of both. When eventually *observed*, the state of a qubit will *collapse* only to the plain states “0” or “1”, with **different probabilities** according to its superposition



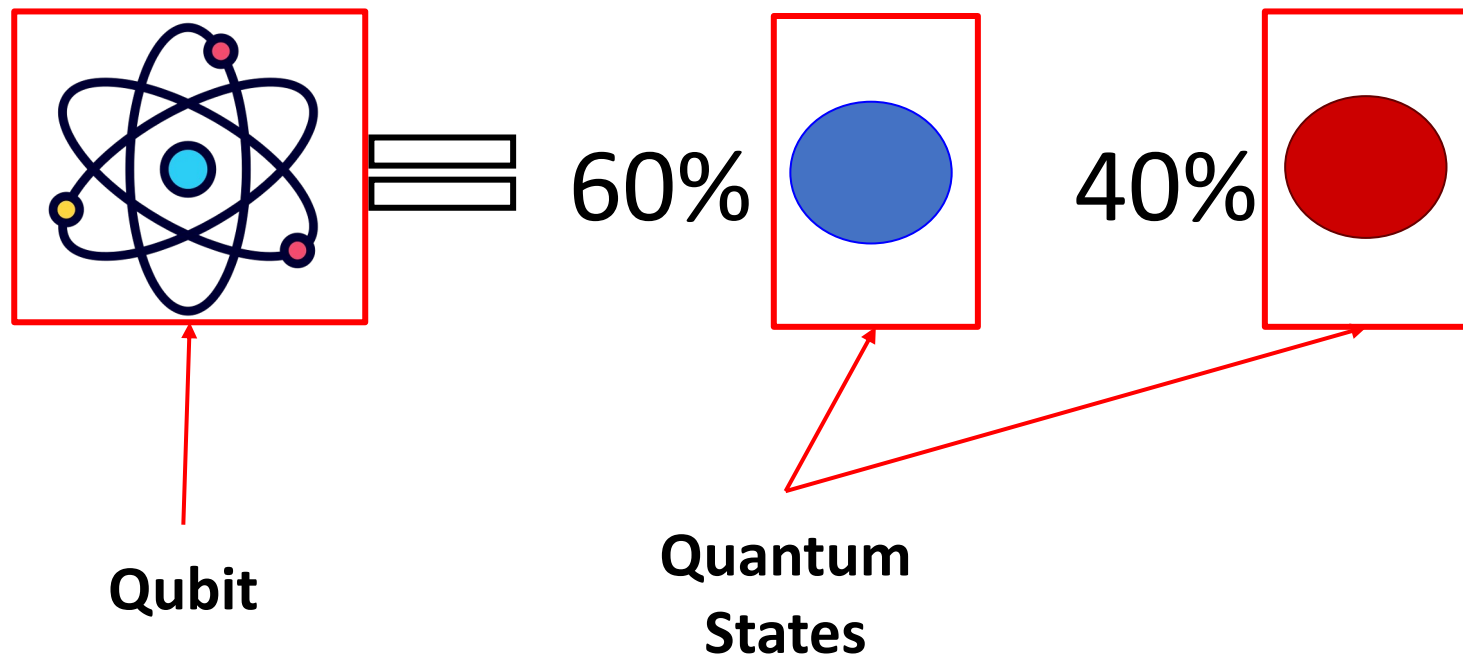
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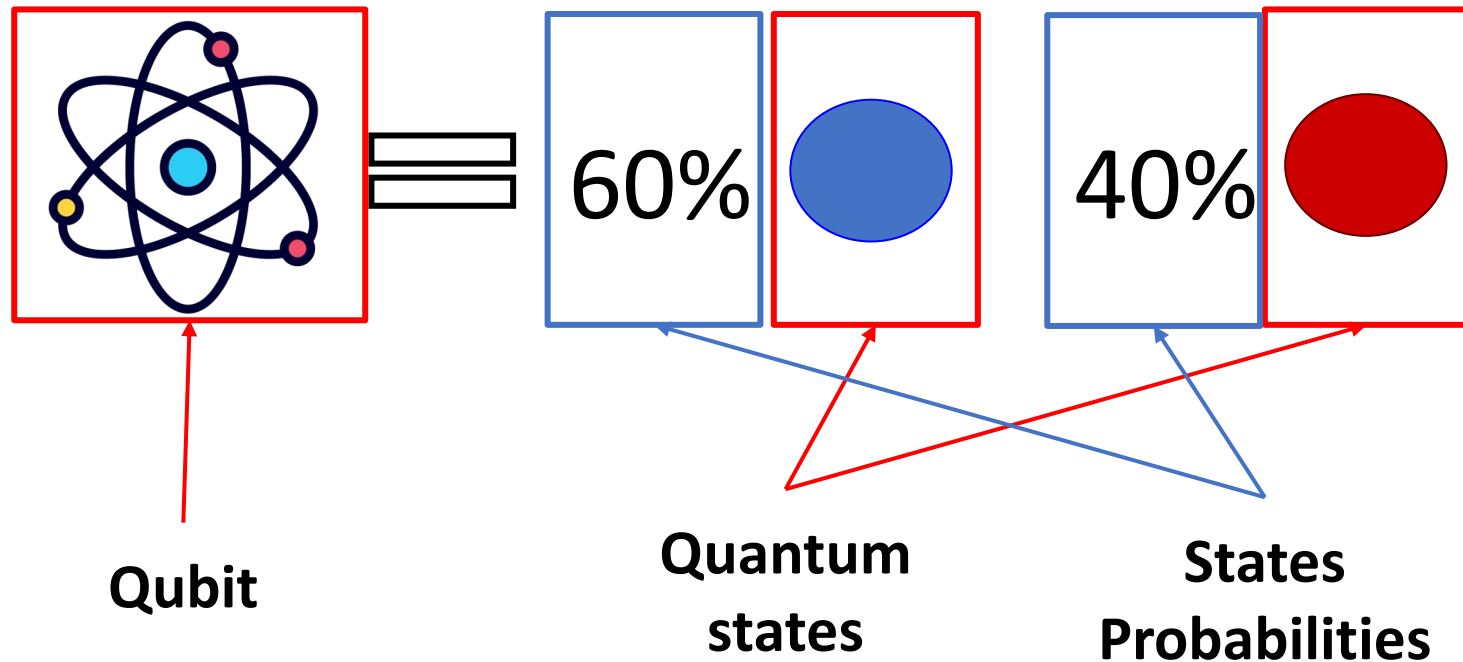
A Qubit

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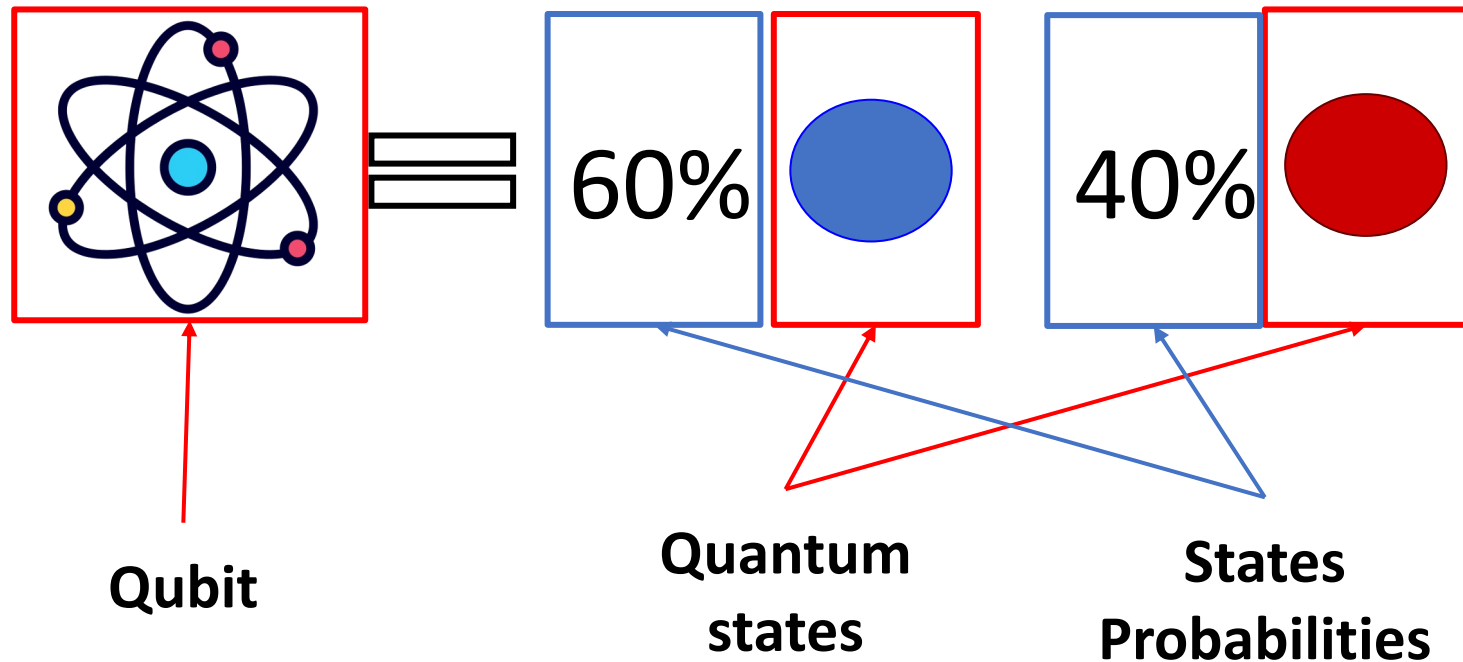
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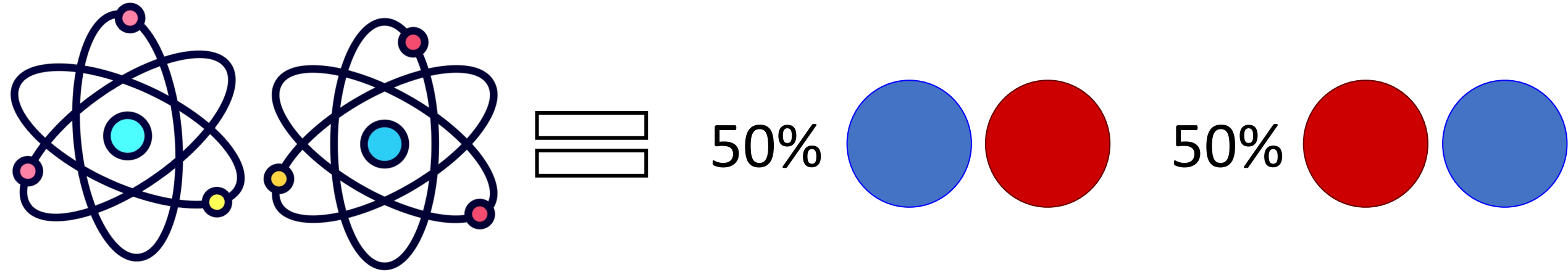
A Qubit

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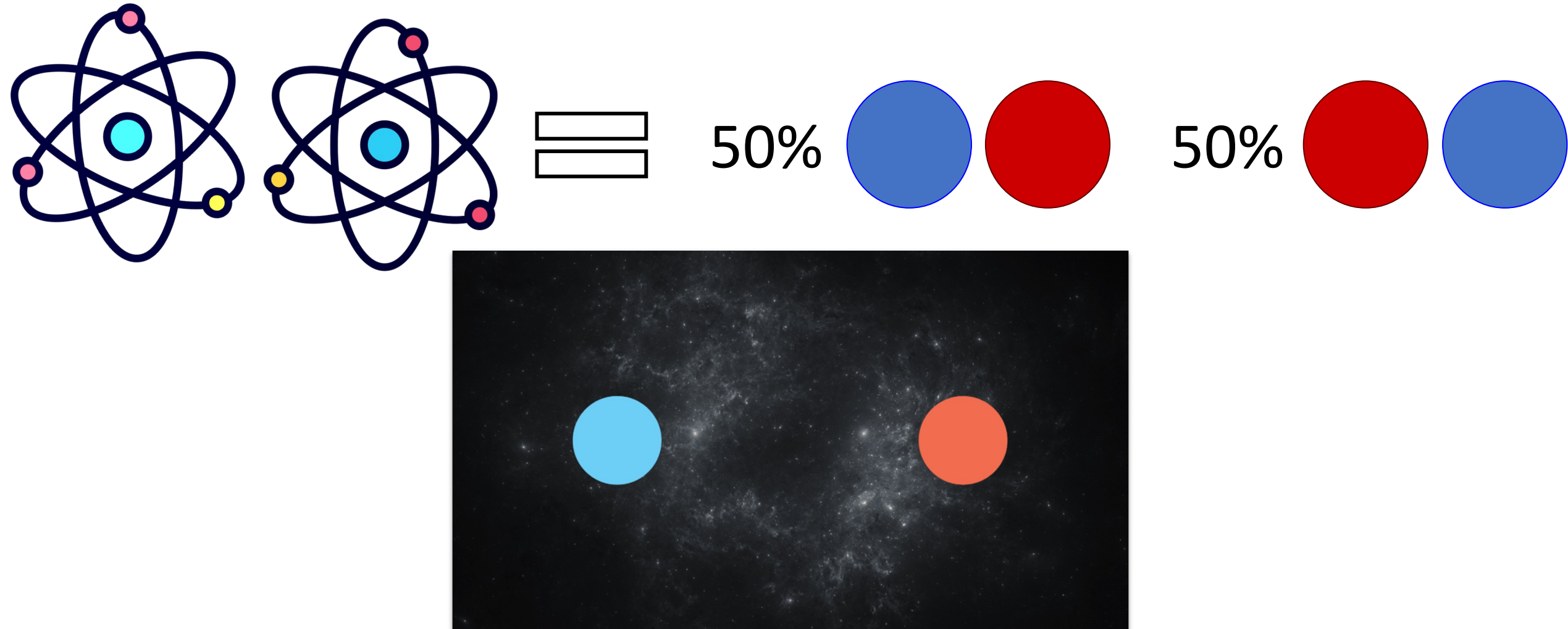
Entanglement

Qubits can be *entangled*, meaning **the state of one qubit** becomes **directly correlated** to **the state of another**, no matter how far apart they are



Entanglement

Qubits can be *entangled*, meaning **the state of one qubit** becomes **directly correlated** to **the state of another**, no matter how far apart they are





Qubit You Choose, Features You Find

Classification	Description	Examples	Qubit lifetime (1)	Gate fidelity (2)	Gate operation time (3)	Connectivity	Scalability	Pros	Cons
Superconducting	Two level system of a superconducting circuit which forms a qubit (a transmon, first developed at Yale)	IBM, Google, Rigetti, Alibaba, Intel, Quantum Circuits	c.50-100µs	c.99.4%	c.10-50ns	Neighbours	Highly scalable (see OQC coaxmon tech)	- Fast gate times - Builds on existing semiconductor industry	- Typically low longevity - Must be kept very cold to work
Ion trap	Single charged ions trapped in magnetic fields. Energy level of its spin comprises the qubits	IonQ; Alpine Quantum Technologies; Honeywell	>1,000s	c.99.9%	c.3-50µs	All-to-all	TBC	- High gate fidelity - Very stable	- Slow operations
Photonics	Qubits made from single particles of light (photons) operating on silicon chips pathways	PsiQuantum, Xanadu	c. 150µs	c. 98.0%	c.1ns	Unknown	Highly scalable (see Psi Quantum)	- Highly scalable - Utilises existing SC industry infrastructure - No temperature requirements	- Nascent technology - Connectivity to be demonstrated
Neutral atoms	Qubits made from individual atoms (rather than ions which have a charge)	Atom Computing, PASQAL, QuEra	Similar to ion trap	c.95%	TBC	TBC	TBC	- Long qubit coherence times	- Must be kept cold - Nascent
Silicon	Artificial atoms made by adding an electron to a small piece of pure silicon and microwaves control the electrons state	Intel, Silicon Quantum Computing	c. 1-10s	c. 99%	c.1-10ns	Neighbours	Expect high scalability	- Stable - Utilises existing semiconductor industry infrastructure	- Must be kept cold - Nascent
Topological qubits	Qubits made from non-Abelian forms of matter	Microsoft (WIP)	Very high	Very high	Unknown	Unknown	Unknown	- Estimated long lifetime and high fidelities	- Existence to be confirmed

Notes: (1) Record coherence time for a single qubit position state; (2) Highest reported fidelity for two qubit gate operations; (3) Speed of gate operations

Sources: Literature review, TQD expert interviews. Special reference to [BCG reports](#), [Science Mag](#) and [NAE report on quantum computing](#).

s = seconds

µs = microsecond (10^{-6} seconds)

ns = nanosecond (10^{-9} seconds)

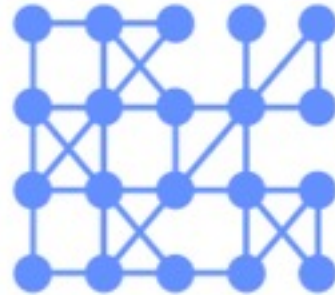


Quantum Topologies

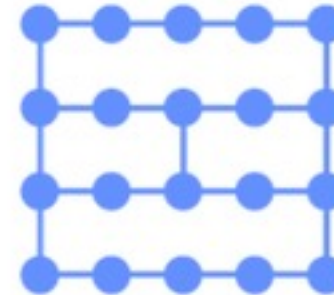
Different qubits should be «physically» connected to be able to perform a multi-qubit operation among such qubits. There are different ways (topologies) we can connect qubits.



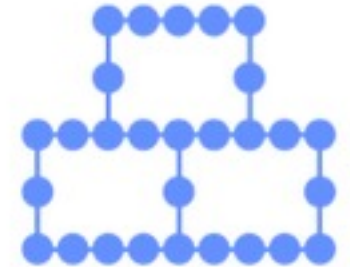
Rueschlikon



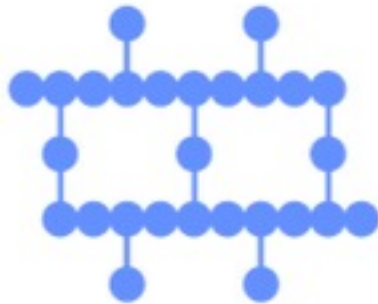
Tokyo



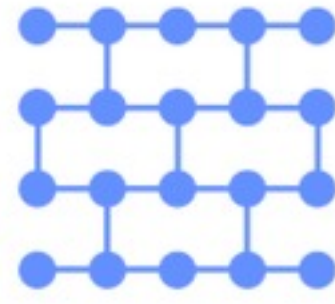
Poughkeepsie
Johannesburg



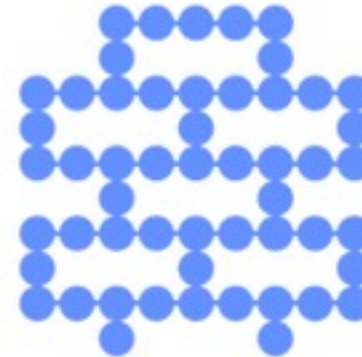
Cambridge



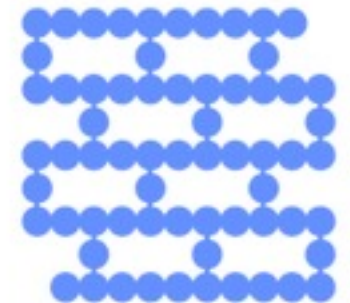
Montreal
Paris
Toronto



Almaden
Singapore
Boebligen

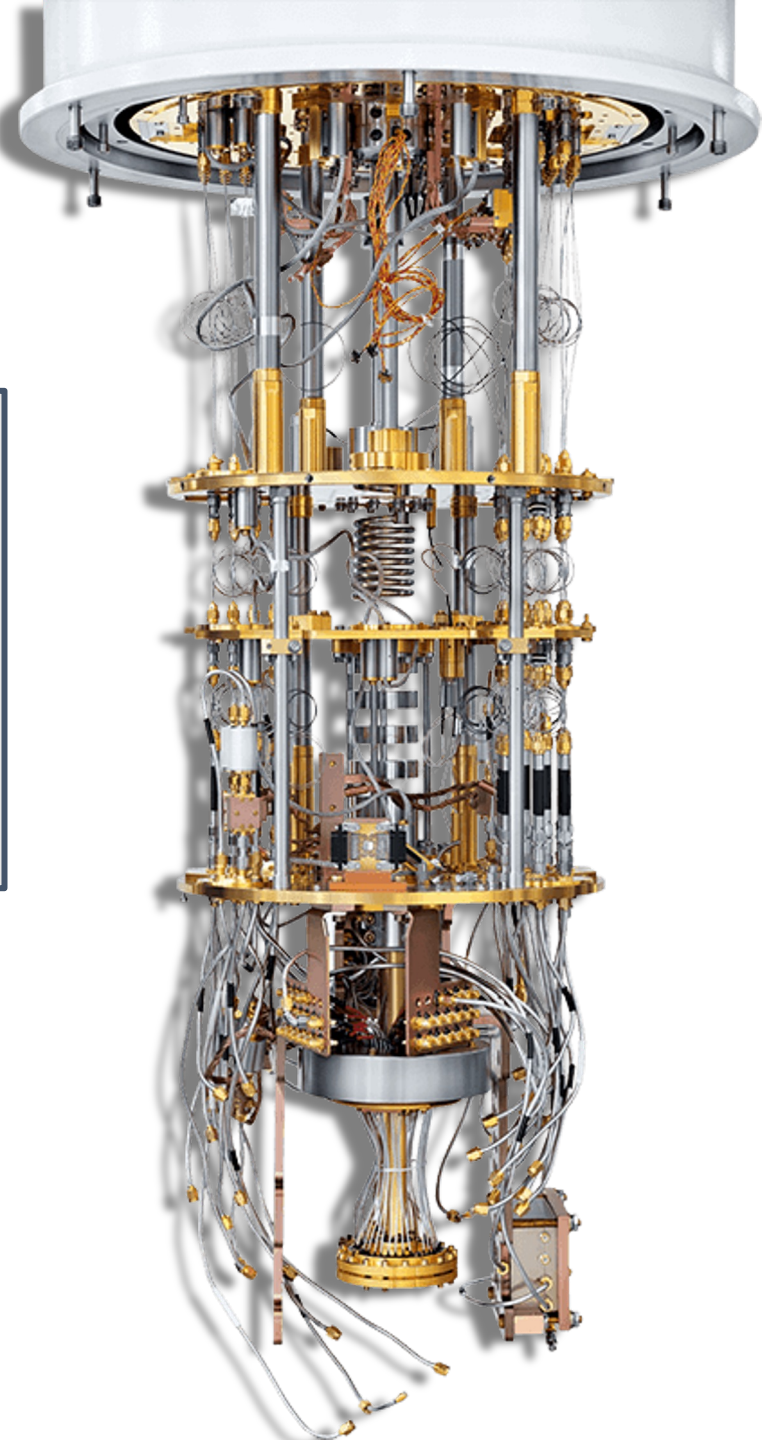
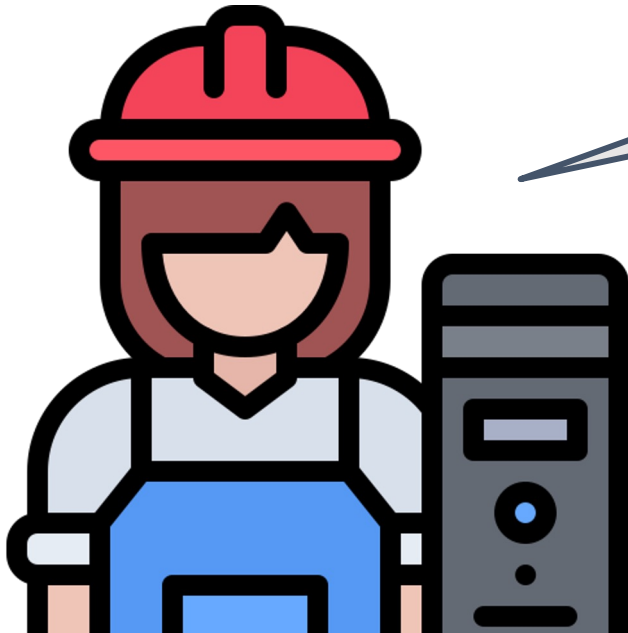


Rochester



Manhattan

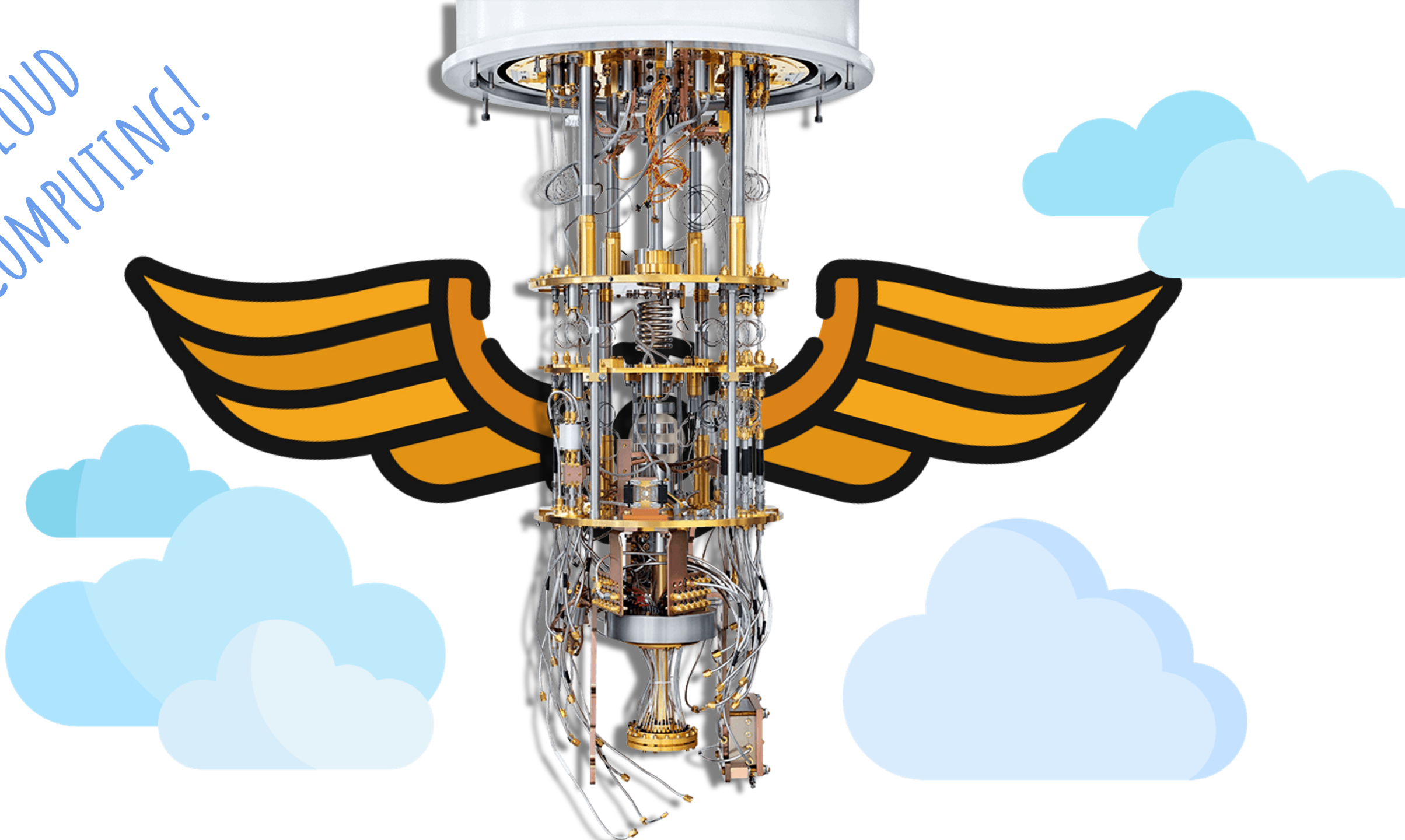
*How can I access a
Quantum Computer?*



IBM Quantum
System One



CLOUD
COMPUTING!



Quantum Computing

Cloud-based Quantum Computing

WELL, ACTUALLY...



Google AI
Quantum

rigetti



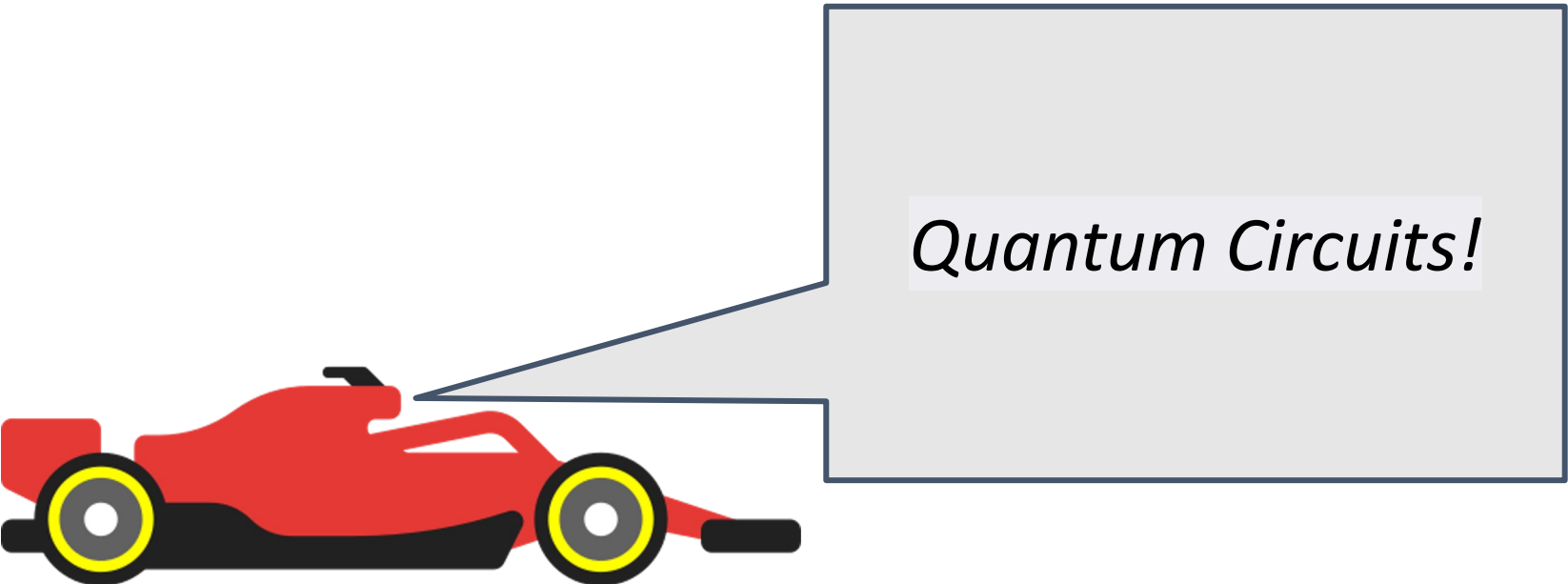
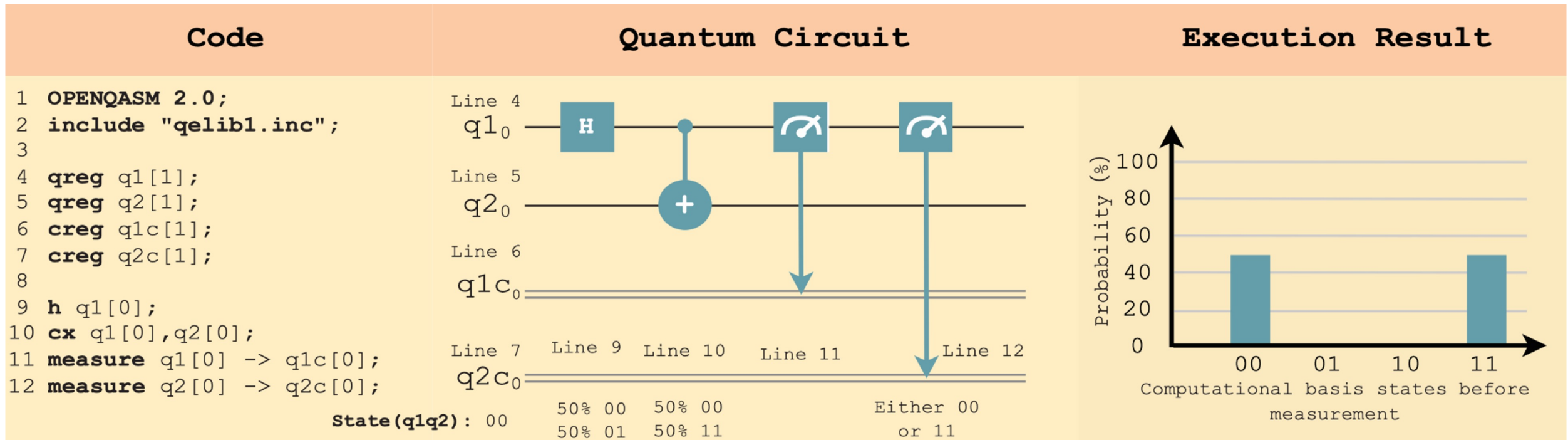
 ibm_cairo		27	64	2.4K	● Online	16	Falcon r5.11	premium	OpenQASM 3
 ibm_auckland	Exploratory	27	64	2.4K	● Online	1818	Falcon r5.11	premium	OpenQASM 3
 ibm_hanoi		27	64	2.3K	● Online - Queue paused	540	Falcon r5.11	premium	OpenQASM 3
 ibm_peekskill	Exploratory	27	-	-	● Online	1	Falcon r8	premium	OpenQASM 3
 ibmq_guadalupe		16	32	2.4K	● Online - Queue paused	37	Falcon r4P	premium	
ibmq_perth		7	32	2.9K	● Online	126	Falcon r5.11H	open	OpenQASM 3
ibmq_lagos		7	32	2.7K	● Online - Reserved	64	Falcon r5.11H	open	OpenQASM 3

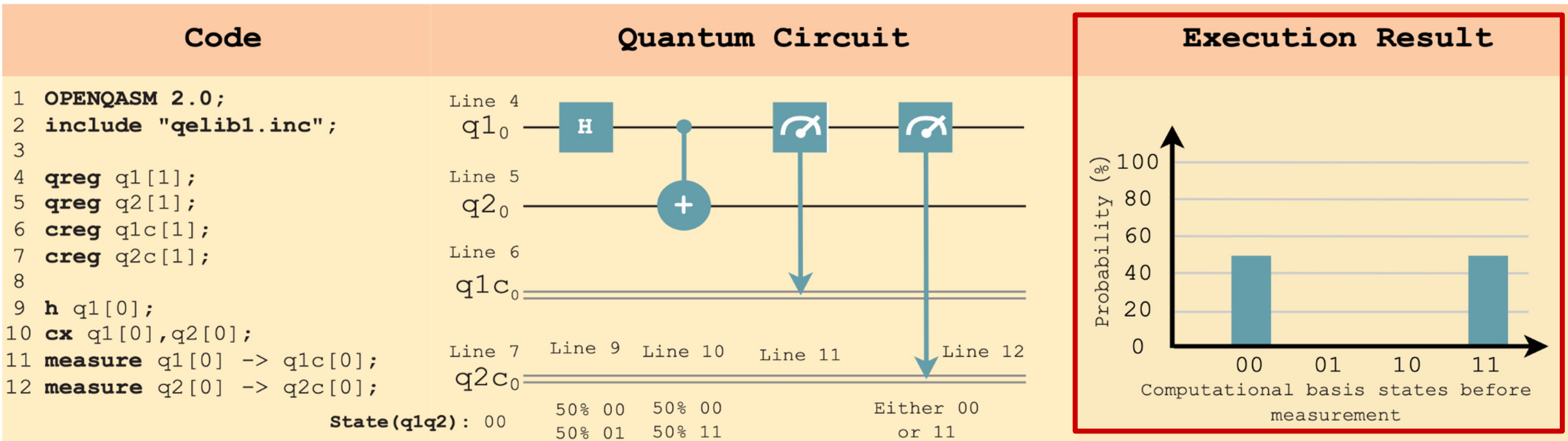
 IonQ IonQ	Quantum Computing	Azure Quantum Credits
 Microsoft Quantum Computing Microsoft	Quantum Computing	Learn & Develop
 Quantinuum Quantinuum	Quantum Computing	Azure Quantum Credits
 Rigetti Quantum Rigetti Computing	Quantum Computing	Azure Quantum Credits

Amazon Web Services	SV1	🟢 AVAILABLE NOW
Amazon Web Services	TN1	🟢 AVAILABLE NOW
Amazon Web Services	DM1	🟢 AVAILABLE NOW
IonQ	Harmony	🕒 02:44:47
IonQ	Aria 1	🕒 02:44:47
Oxford Quantum Circuits	Lucy	🕒 00:44:47
QuEra	Aquila	🕒 1 day 06:44:47
Rigetti	Aspen-M-3	🕒 05:44:47

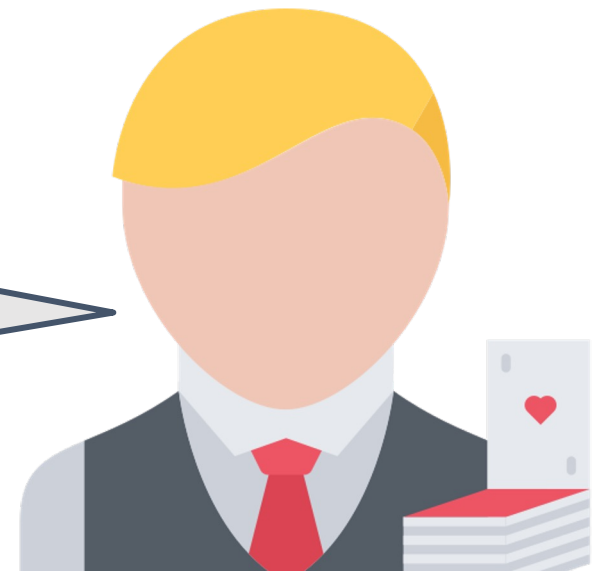
*How I can program a
Quantum Computer?*

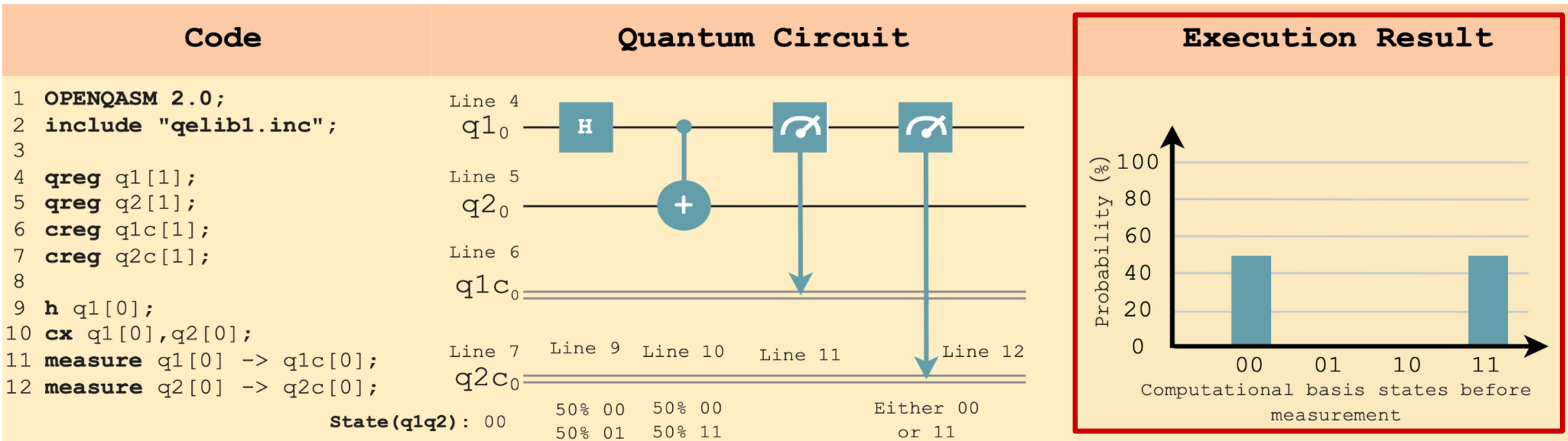




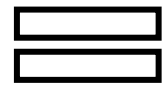
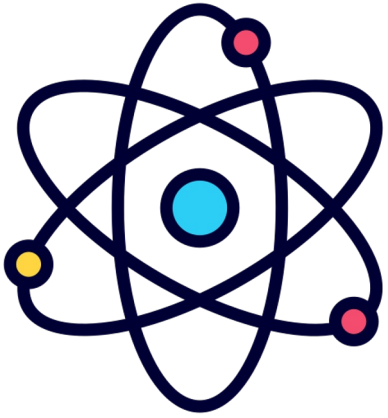


*The output is
probabilistic!*

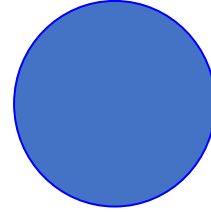




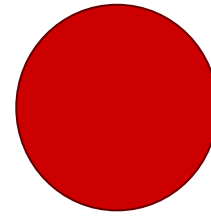
We need multiple iterations (shots)!



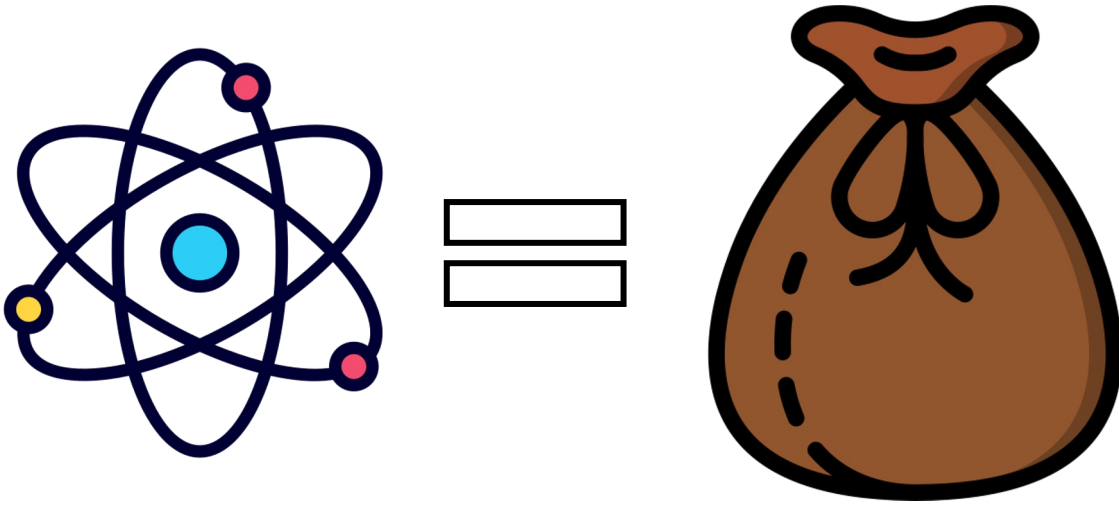
60%



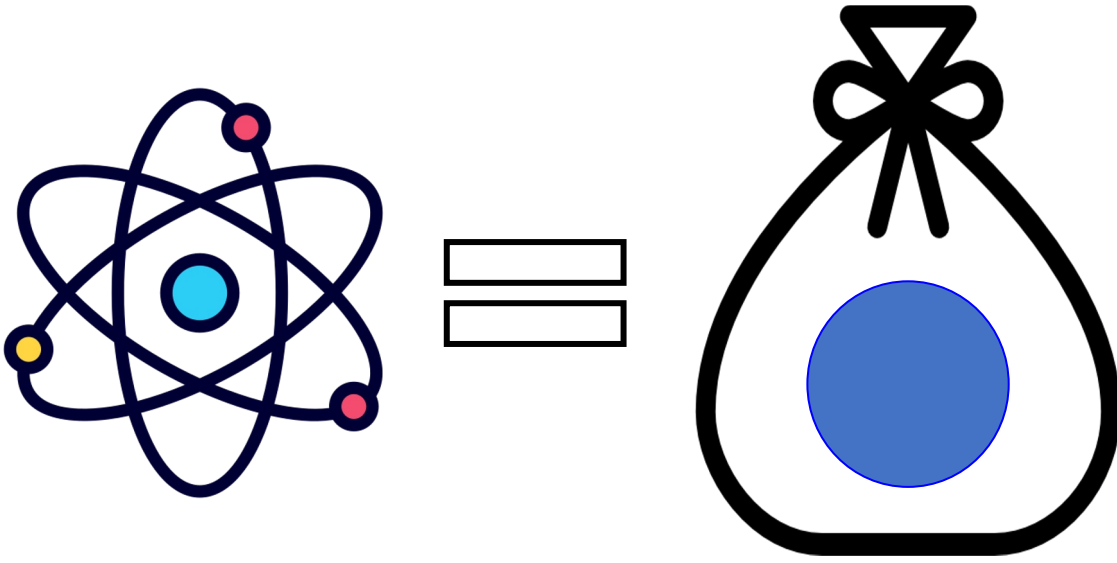
40%



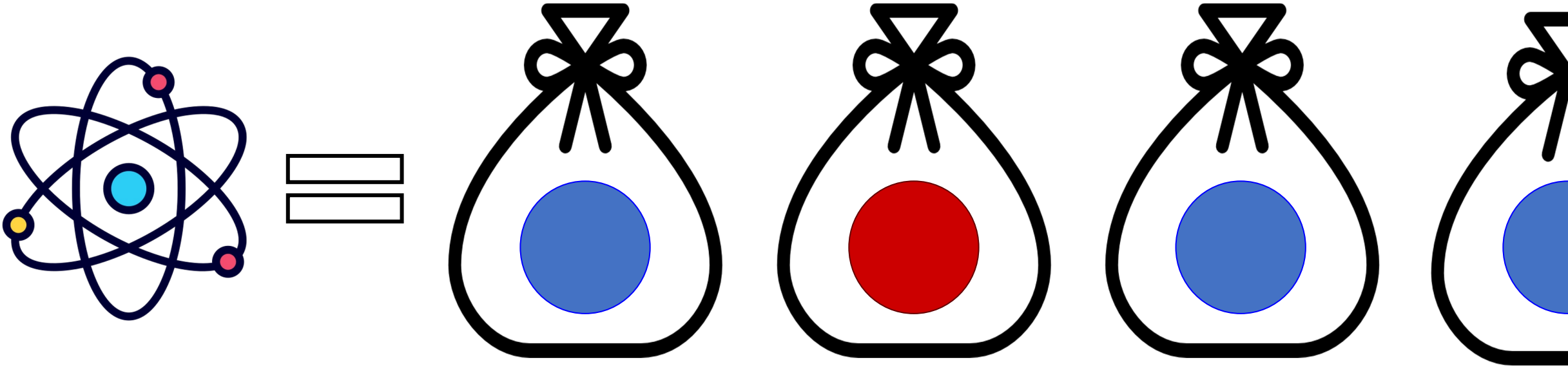
*After the execution of a quantum circuit the quantum state is usually in a **superposition** of multiple final states*



*Unfortunately, laws of physics
prevent us to **directly observe**
the quantum state*



*We can only **measure** the quantum state. When measuring the quantum state, it **collapses** to a single output state*



*We can execute the same circuit
for the same input numerous
times (**shots**) to **approximate**
the **real quantum state**
distribution!*

NISQ Era

Noise
Intermediate
Scale
Quantum

*Currently, we can
execute only small
circuits...*

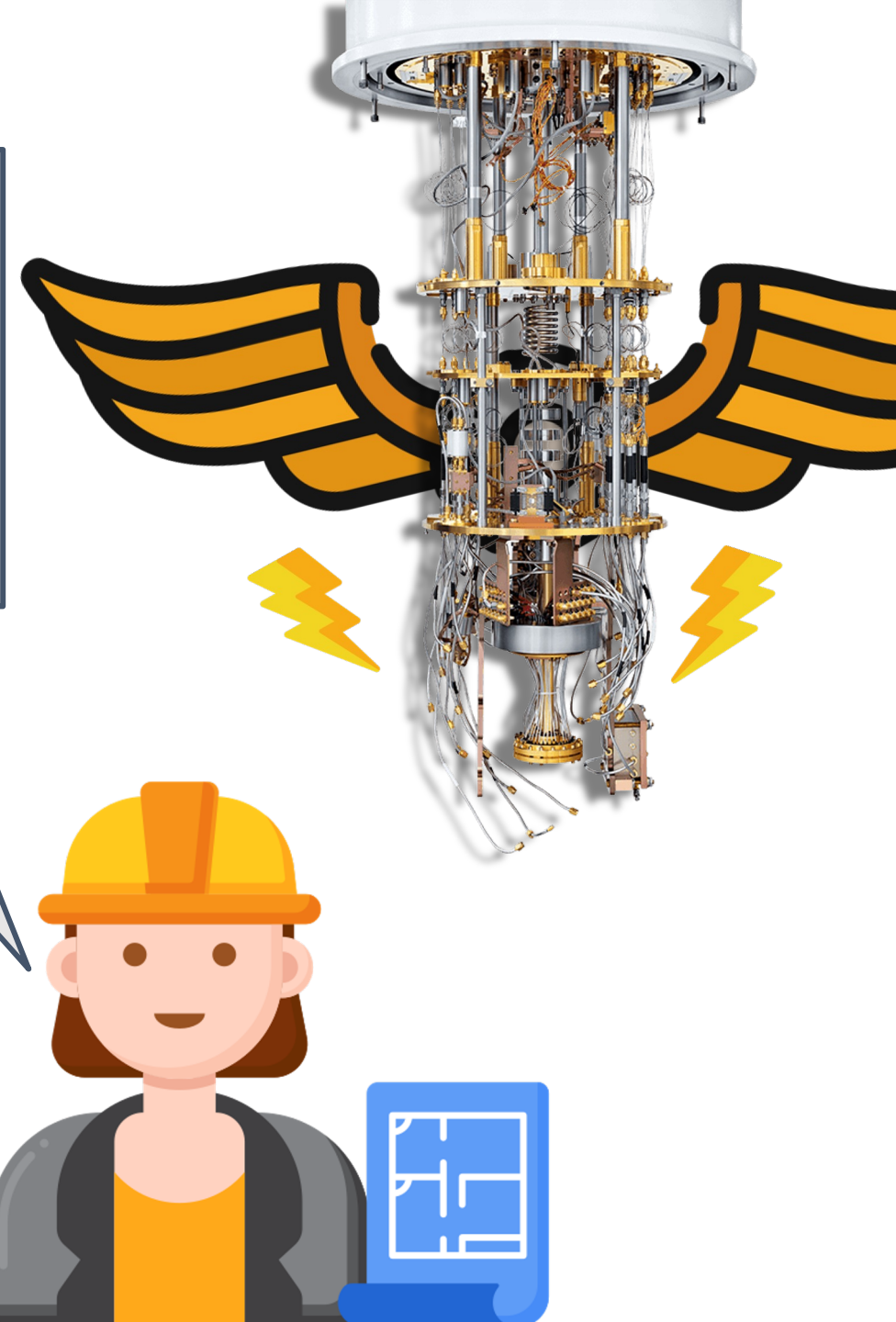
NISQ Devices:

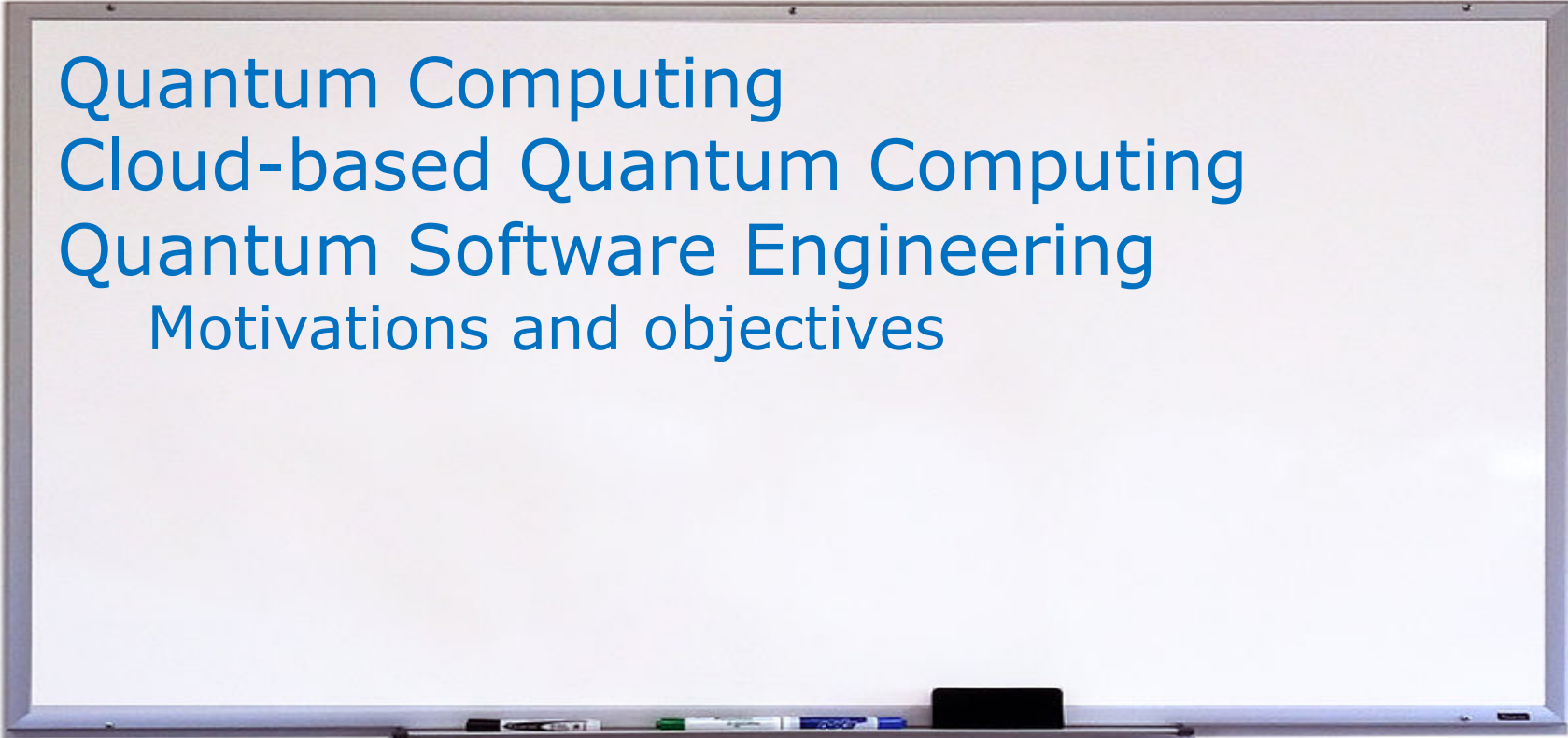
- very **sensitive** to **external interferences** (*noise*)
- **cannot scale** their **size** (*number of qubits*) easily

Consequently:

- can **execute** only *Quantum programs* featuring a **small number** of qubits and **consecutive steps**

Longer programs would be highly affected by noise!



A whiteboard with a silver frame and a white surface. The text is written in blue marker. At the bottom of the whiteboard, there is a black eraser and several markers in black, green, and blue.

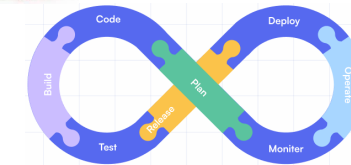
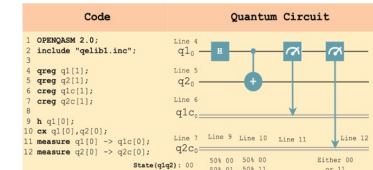
Quantum Computing
Cloud-based Quantum Computing
Quantum Software Engineering
Motivations and objectives

Some current limitations of available quantum computers

- Reliability (error rates)
- Scalability (#qubits)
- Cost

Some current limitations of quantum software development

- Still mainly combining low-level gates into circuits
- Limited software/machine interoperability
- Little support for the whole software lifecycle



Quantum Software Engineering

“The use of **sound engineering principles** for the **development, operation, and maintenance** of quantum software and the associated document to obtain **economically** quantum software that is **reliable** and works **efficiently** on quantum computers”
(Zhao, 2020)

Talavera Manifesto

QSE should

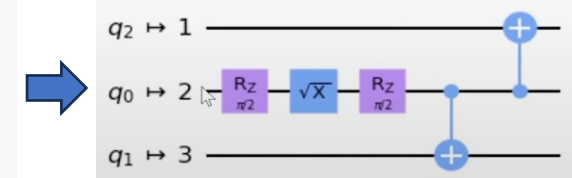
1. be **agnostic** regarding quantum programming languages and technologies
2. embrace the **coexistence** of classical and quantum computing
3. support the **management** of quantum software development projects
4. consider the **evolution** of quantum software
5. aim at delivering quantum programs with desirable **zero defects**
6. assure the **quality** of quantum software
7. promote quantum software **reuse**
8. address **security** and **privacy** by design
9. cover the **governance** and **management** of software

(M. Piattini et al., 2020)

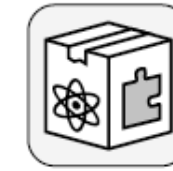
Quantum Software Engineering: examples

Improving **transpilers**

```
qc = QuantumCircuit(3)
qc.h(0)
qc.cx(0, 1)
qc.cx(0, 2)
qc.draw(output='mpl')
```



Enabling **code reusability**



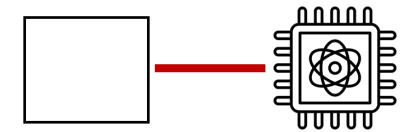
Q Module Template

Enabling **program verification**

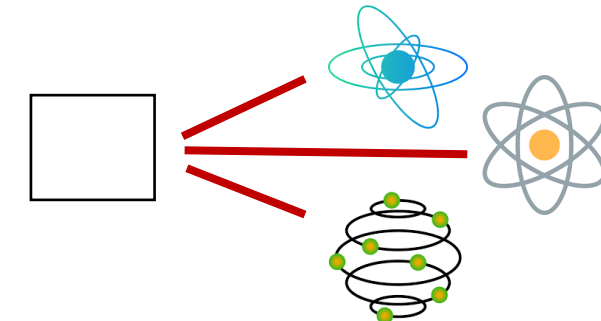
$$P \sim Q \quad P \preceq Q$$

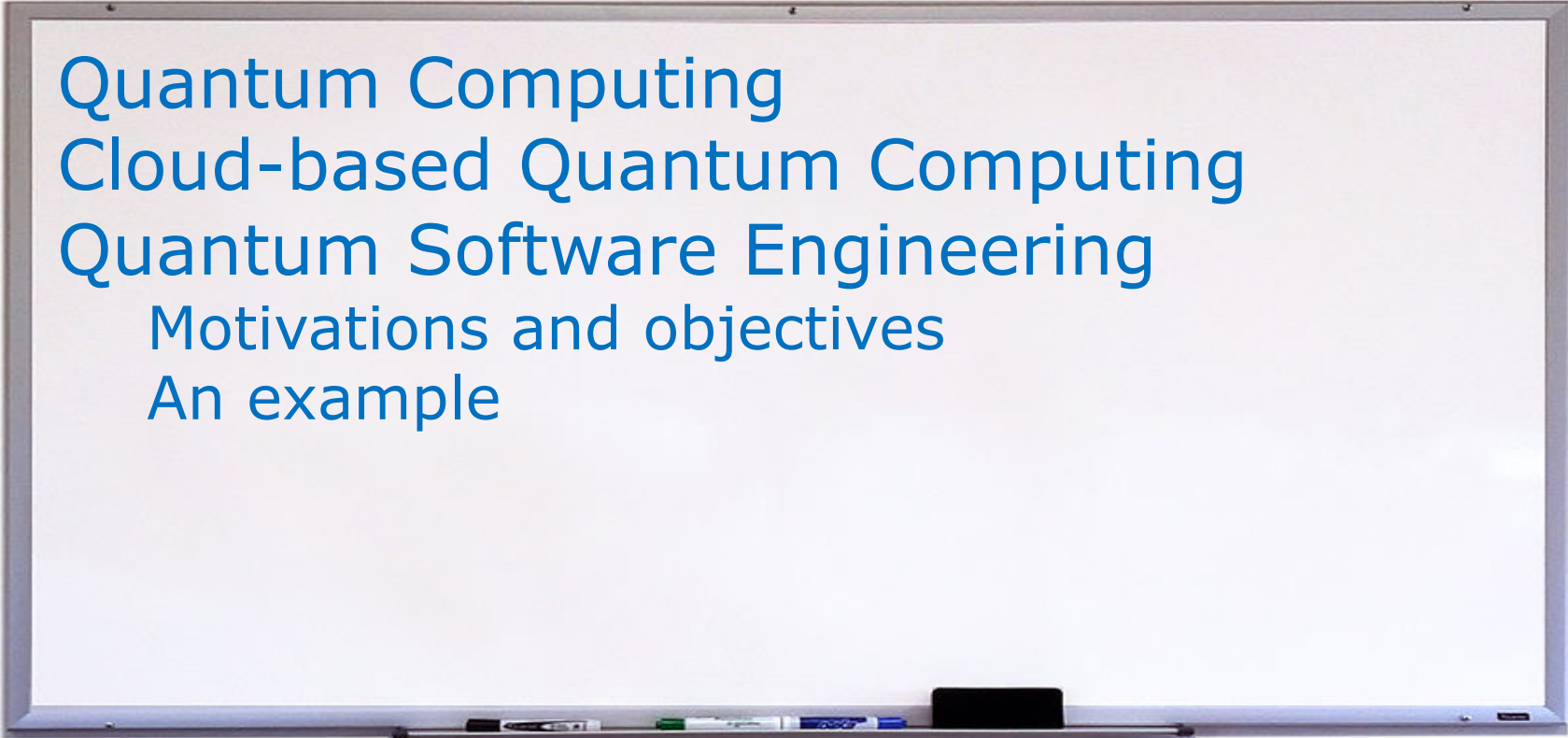
Enabling the **coexistence** of classical & quantum computations

$$\mathbf{x} = \mathbf{f}(\mathbf{y}) \oplus \mathbf{q}(\mathbf{z})$$



Better exploiting **heterogeneous unreliable** Quantum computers





Quantum Computing
Cloud-based Quantum Computing
Quantum Software Engineering
 Motivations and objectives
 An example

Code

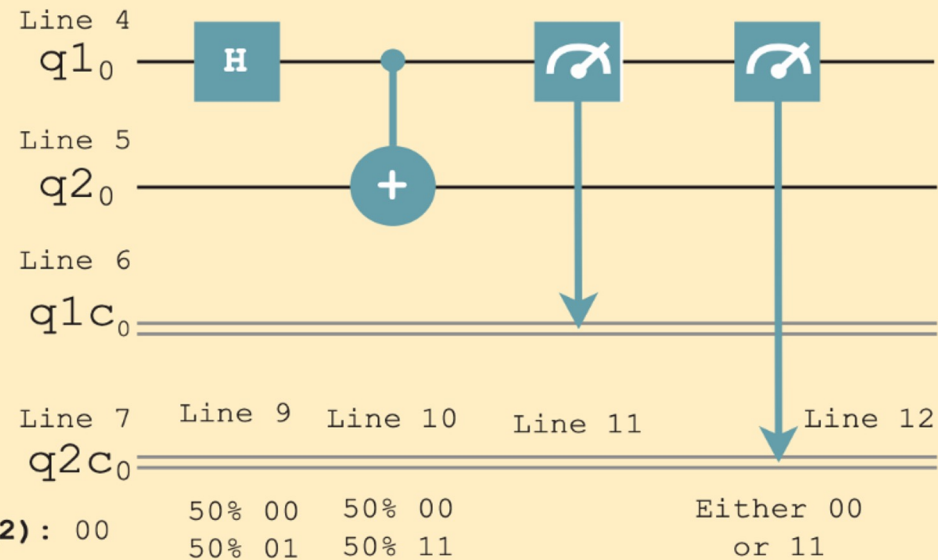
```

1 OPENQASM 2.0;
2 include "qelib1.inc";
3
4 qreg q1[1];
5 qreg q2[1];
6 creg q1c[1];
7 creg q2c[1];
8
9 h q1[0];
10 cx q1[0],q2[0];
11 measure q1[0] -> q1c[0];
12 measure q2[0] -> q2c[0];

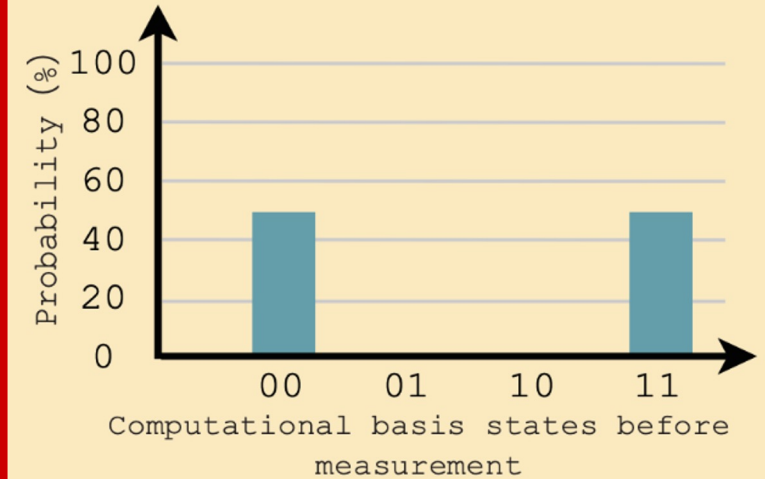
```

State(q1q2): 00

Quantum Circuit



Execution Result



*The output of a quantum computation is **probabilistic!***

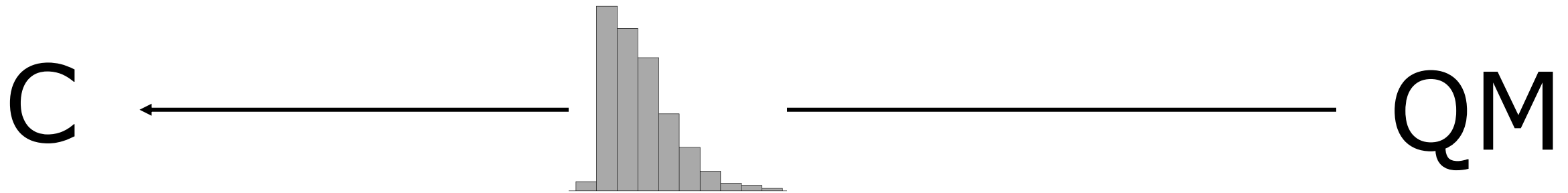


*We need multiple iterations (**shots**)!*

Executing a circuit on a single QM



Executing a circuit on a single QM



Introducing a broker

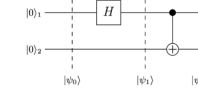
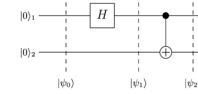
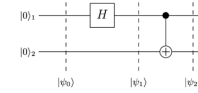
C

B

QM₁

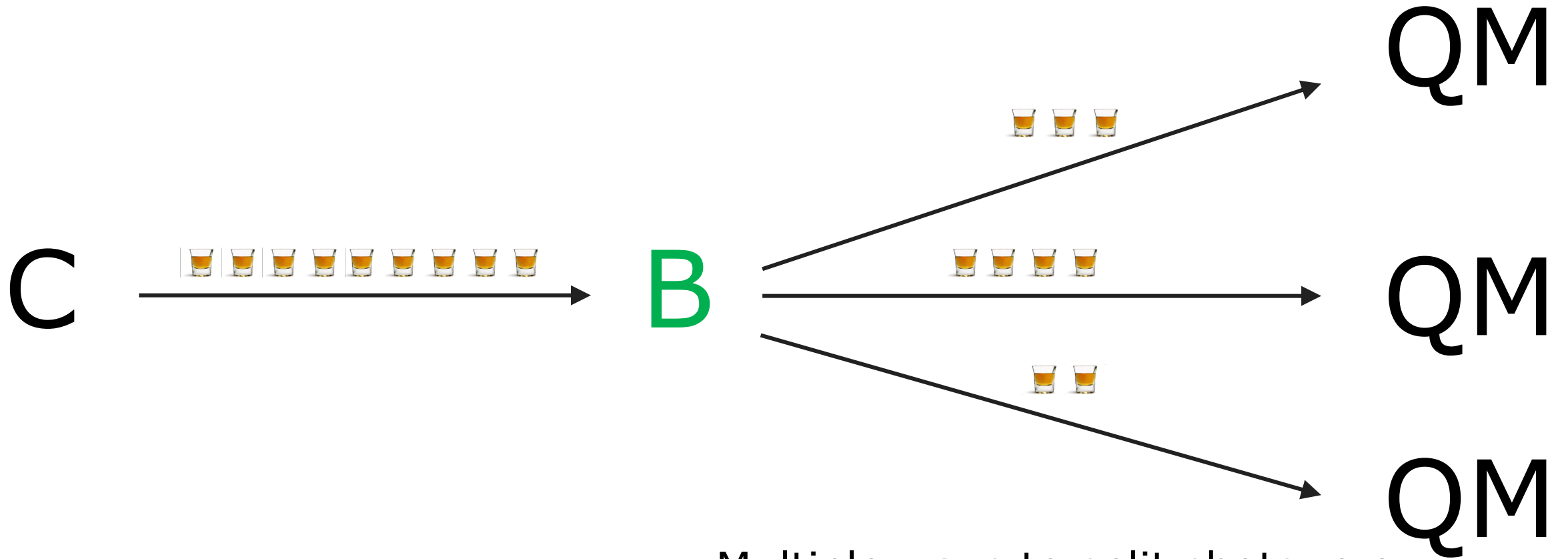
QM₂

QM₃



Calibration: executing a set of simple circuits to estimate reliability of each QM_{*i*}

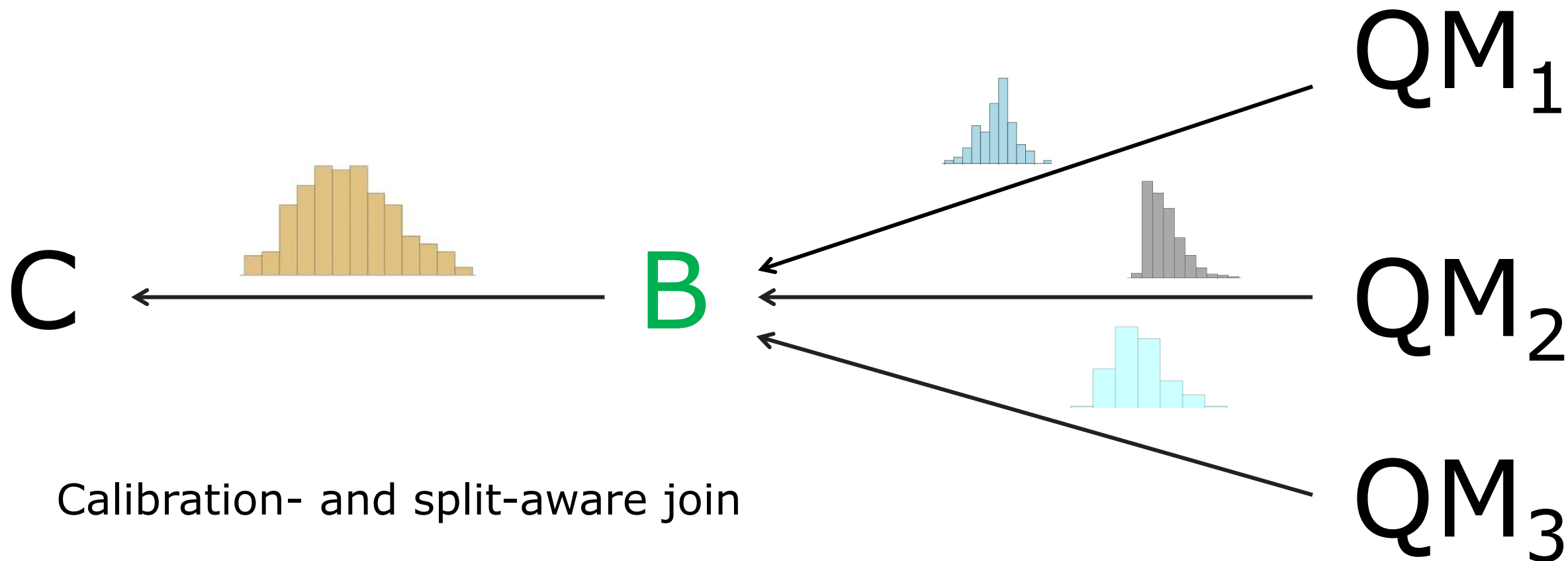
Splitting shots



Multiple ways to split shots, e.g.

- Uniform
- Random
- Noise-aware
- Cost-aware

Joining results



Why splitting shots across different QMs?

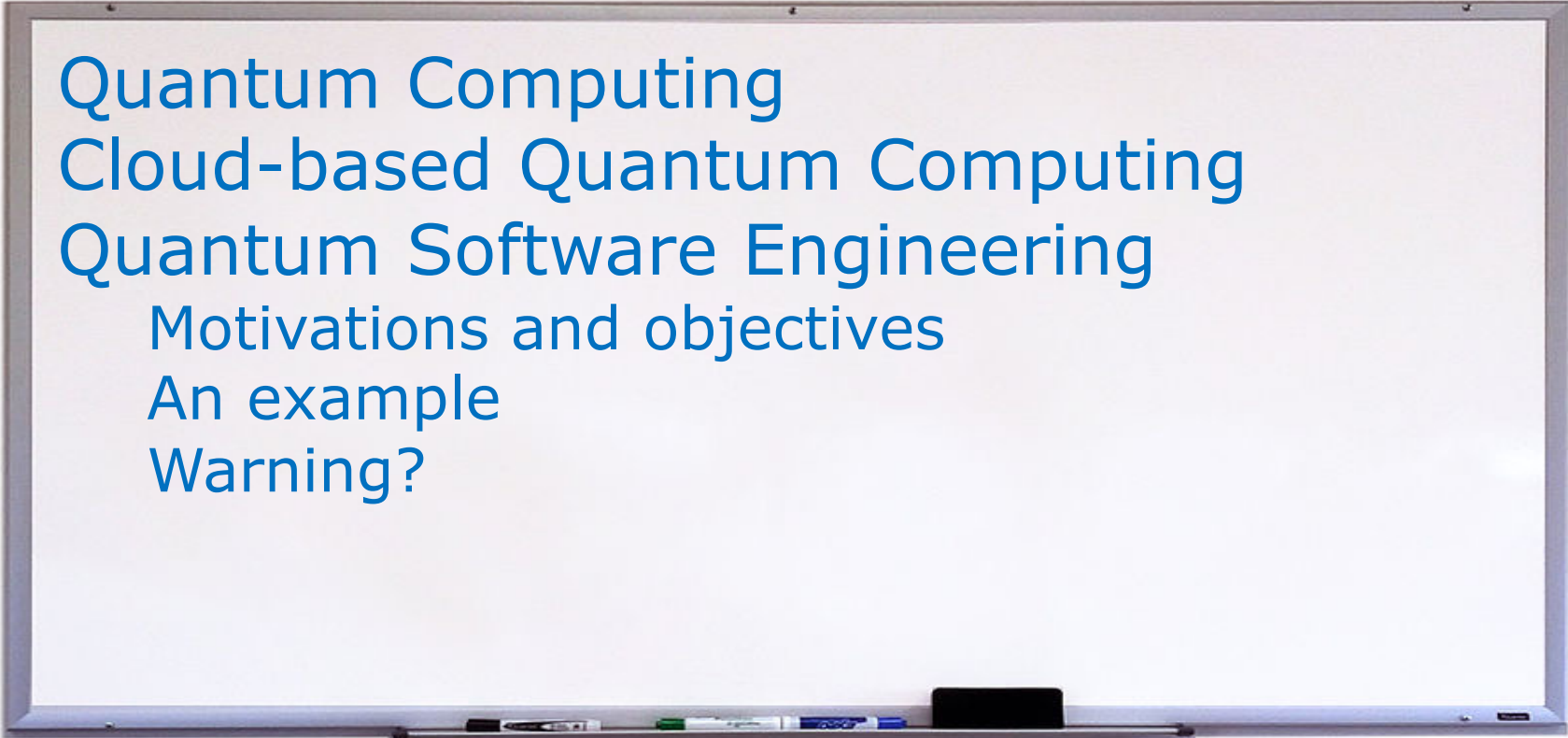
1 + 1 + 1 > 3 ?

Yes!

reliability

cost

better results



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 Warning?

QSE: warning?



QSE: warning?

